EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

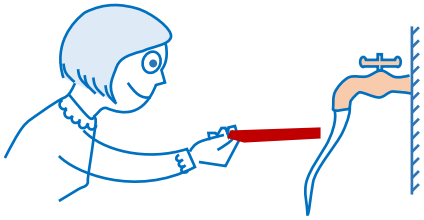
Properties of Charge and Coulomb's Law

1. If an object made of substance *A* rubs an object made of substance *B*, then *A* becomes positively charged and *B* becomes negatively charged. If, however, an object made of substance *A* is rubbed against an object made of substance *C*, then *A* becomes negatively charged. What will happen if an object made of substance *B* is rubbed against an object made of substance *C*?

(A) *B* becomes positively charged and *C* becomes positively charged.
(B) *B* becomes positively charged and *C* becomes negatively charged.
(C) *B* becomes negatively charged and *C* becomes positively charged.
(D) *B* becomes negatively charged and *C* becomes negatively charged.

PEO076

2. In normal cases thin stream of water bends toward a negatively charged rod. When a positively charged rod is placed near the stream, it will bend in the



(A) Opposite direction.
(B) Same direction.
(C) It won't bend at all.
(D) Can't be predicted.

PEO077

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3. Two charged bodies *A* and *B* exert repulsive forces on each other. If charge on *A* is more than that on *B*, which of the following statement is true.

(A) Body *A* experiences more Colombian force then *B*.
(B) Body *A* experiences less Colombian force then *B*.
(C) Both of them experience Colombian forces of equal magnitude.
(D) It depends whether the bodies can be treated as point like charges or not.

PEO078

4. Given are four arrangements of three fixed electric charges. In each arrangement, a point labelled *P* is also identified — test charge, $+q$, is placed at point *P*. All of the charges are of the same magnitude *Q*, but they can be either positive or negative as indicated. The charges and point *P* all lie on a straight line. The distances between adjacent items, either between two charges or between a charge and point *P*, are all the same. Correct order of choices in a decreasing order of magnitude of force on *P* is

I. (+) (+) (+) P II. (+) (+) P (−) III. (+) (+) (−) P IV. (+) (−) (+) P

(A) II > I > III > IV
(B) I > II > III > IV
(C) II > I > IV > III
(D) III > IV > I > II

PEO079

5. Two particle of equal mass *m* and charge *q* are placed at a distance of 16 *cm*. They do not experience any force. The value of $\frac{q}{m}$ is

(A) *I*
(B) $\sqrt{\frac{\pi\varepsilon_0}{G}}$
(C) $\sqrt{\frac{G}{4\pi\varepsilon_0}}$
(D) $\sqrt{4\pi\varepsilon_0 G}$

PEO082

For More Join: @IITJEE_Advanced

6. Two equally charged, identical metal spheres A and B repel each other with a force 'F'. The spheres are kept fixed with a distance 'r' between them. A third identical, but uncharged sphere C is brought in contact with A and then placed at the mid-point of the line joining A and B. The magnitude of the net electric force on C is

(A) F (B) $3F/4$ (C) $F/2$ (D) $F/4$

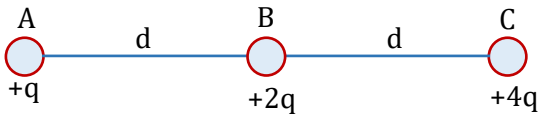
PEO083

7. Two point charges placed at a certain distance r in air exert a force F on each other. Then the distance r' at which these charges will exert the same force in a medium of dielectric constant k is given by

(A) r (B) r/k (C) $r/\sqrt{k}$ (D) $r\sqrt{k}$

PEO085

8. Three charges $+q$, $+2q$ and $+4q$ are connected by strings as shown in the figure. What is ratio of tensions in the strings AB and BC.



(A) 1 : 2 (B) 1 : 3 (C) 2 : 1 (D) 3 : 1

PEO086

9. A charge q is placed at the center of the line joining two equal charges Q. The system of the three charges will be in equilibrium if q is equal to

(A) $-Q/2$ (B) $-Q/4$ (C) $+Q/4$ (D) $+Q/2$

PEO093

10. Two identical charges, 5 μC each, are in the air. What is the largest charge to be transferred from one charge to another, so that their interaction force decreases 2.5 times ?

(A) $\sqrt{11}\ \mu C$ (B) $4\ \mu C$ (C) $\sqrt{13}\ \mu C$ (D) $\sqrt{15}\ \mu C$

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PEO094

11. A point charge $-3q$ is positioned at point (2, −1, 3) of a Cartesian coordinate system. A test point charge $2q$ is placed at point (3, 0, 4). Which of the following is the unit vector in the direction of the force acting on the test charge? All the coordinates are in meters.

(A) $\hat{i}+\hat{j}+\hat{k}$ (B) $-(\hat{i}+\hat{j}+\hat{k})$ (C) $(\hat{i}+\hat{j}+\hat{k})/\sqrt{3}$ (D) $-(\hat{i}+\hat{j}+\hat{k})/\sqrt{3}$

PEO090

12. Two-point charge of 100 μC and -4 μC are positioned at points $(-2\sqrt{3}, 3\sqrt{3}, -4)$ and $(4\sqrt{3}, -5\sqrt{3}, 6)$ respectively of a Cartesian coordinate system. Find the force vector on the -4 μC charge? All the coordinates are in meters.

(A) $9\times10^{-4}\left(3\sqrt{3}\hat{i}-4\sqrt{3}\hat{j}+5\hat{k}\right)$ (B) $9\times10^{-4}\left(-3\sqrt{3}\hat{i}+4\sqrt{3}\hat{j}-5\hat{k}\right)$

(C) $2.25\times10^{-4}\left(-3\sqrt{3}\hat{i}+4\sqrt{3}\hat{j}-5\hat{k}\right)$ (D) $2.25\times10^{-4}\left(3\sqrt{3}\hat{i}-4\sqrt{3}\hat{j}+5\hat{k}\right)$

PEO095

Electric Field

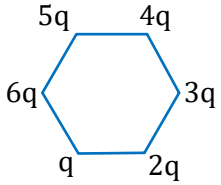
13. There are two point charges q_1 and q_2 lying on a circle of unit radius. Electric field intensity at the center of circle due to these charges is $\vec{E}$. Find the position vector of the center with respect to q_2 if the position vector of the center with respect to q_1 is $\vec{r}_1$.

(A) $\dfrac{\vec{E}+kq_1\vec{r}_1}{kq_2}$ (B) $-\left(\dfrac{\vec{E}+kq_1\vec{r}_1}{kq_2}\right)$ (C) $\dfrac{kq_1\vec{r}_1-\vec{E}}{kq_2}$ (D) $\dfrac{\vec{E}-kq_1\vec{r}_1}{kq_2}$

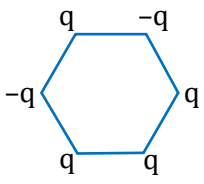
PEO097

For More Join: @IITJEE_Advanced

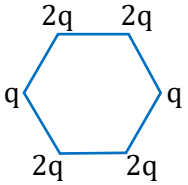
14. Figure below shows regular hexagon, with different charges placed at the vertices. In which of the following cases is the electric field at the centre zero?



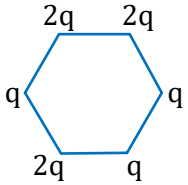
I



II



III



IV

(A) IV (B) III (C) I (D) II

PEO116

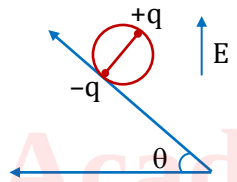
15. Two point charges 3μC and 2.5 μC are placed at point A (1, 1, 2)m and B (0, 3, −1)m respectively. Find out electric field intensity vector at point C(3, 3, 3)m.

(A) $[2500\,\hat{i}+2000\,\hat{j}+1720\hat{k}\,]\ N/C$ (B) $[2000\,\hat{i}+2000\,\hat{j}+1720\hat{k}\,]\ N/C$

(C) $[1720\,\hat{i}+2000\,\hat{j}+1720\hat{k}\,]\ N/C$ (D) $[2540\,\hat{i}+2000\,\hat{j}+1720\hat{k}\,]\ N/C$

PEO117

16. A wheel having mass m has charges $+q$ and $-q$ on diametrically opposite points. It remains in equilibrium on a rough inclined plane in the presence of uniform vertical electric field E =

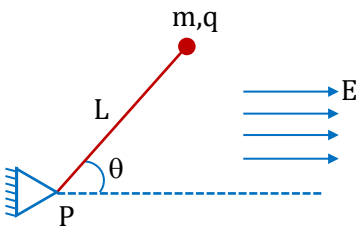


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(A) $\frac{mg}{q}$ (B) $\frac{mg}{2q}$ (C) $\frac{mg\tan\theta}{2q}$ (D) None

PEO100

17. A particle of mass m and charge q is attached to a light rod of length L. The rod can rotate freely in the plane of paper about the other end, which is hinged at P. The entire assembly lies in a uniform electric field E also acting in the plane of paper as shown. The rod is released from rest when it makes an angle θ with the electric field direction. Determine the speed of the particle when the rod is parallel to the electric field.



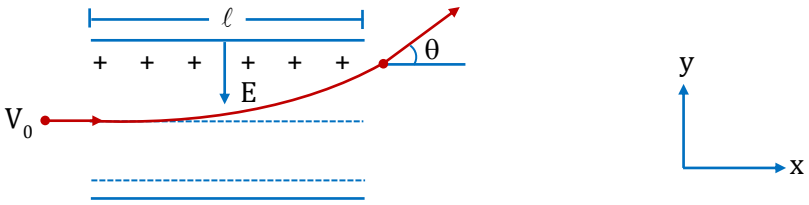
(A) $\left(\frac{2qEL(1-\cos\theta)}{m}\right)^{1/2}$ (B) $\left(\frac{2qEL(1-\sin\theta)}{m}\right)^{1/2}$

(C) $\left(\frac{qEL(1-\cos\theta)}{2m}\right)^{1/2}$ (D) $\left(\frac{2qEL\cos\theta}{m}\right)^{1/2}$

PEO110

For More Join: @IITJEE_Advanced

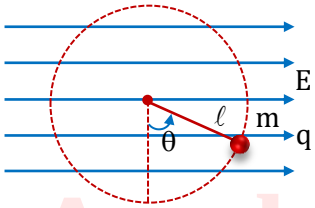
18. A uniform electric field $E = 91 \times 10^{-6}\ V/m$ is created between two parallel, charged plates as shown in figure. An electron enters the field symmetrically between the plates with a speed $v_0 = 4\times10^3\ m/s$. The length of each plate is $\ell = 1m$. Find the angle of deviation of the path of the electron as it comes out of the field. (Mass of the electron is $m = 9.1 \times 10^{-31}\ kg$ and its charge is $e = -1.6 \times 10^{-19}\ C$).



(A) $\theta = 30°$ (B) $\theta = 15°$ (C) $\theta = 45°$ (D) $\theta = 60°$

PEO112

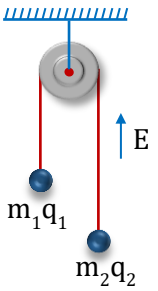
19. A small ball of mass m and charge $+q$ tied with a string of length l, is rotating in a vertical circle under gravity and a uniform horizontal electric field E as shown. The tension in the string will be minimum for:-



(A) $\theta = \tan^{-1}\left(\frac{qE}{mg}\right)$ (B) $\theta = \pi$ (C) $\theta = \pi - \tan^{-1}\left(\frac{qE}{mg}\right)$ (D) $\theta = \pi + \tan^{-1}\left(\frac{qE}{mg}\right)$

PEO099

20. Two small balls are carrying masses and charges m_1, q_1 and m_2, q_2 respectively. The entire system is placed in uniform electric field $\vec{E}$ directed upward. If $(m_1 + m_2)\ g = (q_1 + q_2)E$, then the acceleration of centre of mass. (Neglect the interaction between balls).

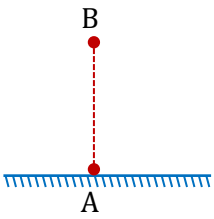


(A) is necessarily zero (B) is not necessarily zero

(C) is directed downward (D) none

PEO105

21. Particle B of charge Q and mass m is in equilibrium under weight and electrostatics force applied by a fixed charged A, which is directly beneath the particle B as shown in figure. When particle B is disturbed along vertical, the equilibrium is



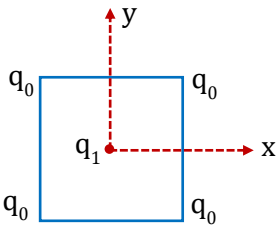
(A) stable (B) unstable

(C) neutral (D) there can not be in equilibrium

PEO098

For More Join: @IITJEE_Advanced

22. Four equal charges q_0 each, are kept fixed on the vertices of a square of side 'a'. Charge q_1 is kept at the center of the square as shown in the figure. Then which statement is correct ?



(Here charge q_1 is restricted to move perpendicular to XY plane i.e. along z-axis)

(A) charge q_1 is always in stable equilibrium.

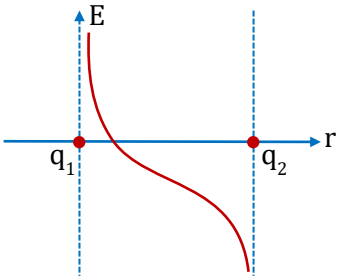
(B) charge q_1 is in unstable equilibrium.

(C) charge q_1 is always in stable equilibrium only if q_1 and q_0 have different (opposite) signs.

(D) charge q_1 is always in stable equilibrium only if q_1 and q_0 have same signs.

PEO103

23. The variation of electric field between the two charges q_1 and q_2 along the line joining the charges is plotted against distance from q_1 (taking rightward direction of electric field as positive) as shown in the figure. Then the correct statement is :-



(A) q_1 and q_2 are positive and $q_1 < q_2$

(B) q_1 and q_2 are positive and $q_1 > q_2$

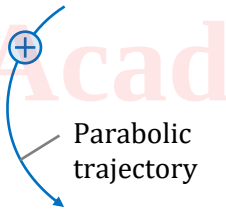
(C) q_1 is positive and q_2 is negative and $q_1 < q_2$

(D) q_1 and q_2 are negative and $q_1 < q_2$

PEO109

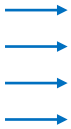
24. Which electric field is responsible for the proton's trajectory as shown below ?

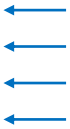
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(A) 

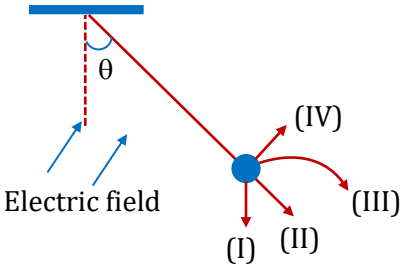
(B) 

(C) 

(D) 

PEO113

25. A small object of mass M and charge Q is connected to an insulating massless string in vacuum on Earth. A uniform electric field exists throughout the region of the vacuum as indicated. The mass remains in static equilibrium at an angle of θ with the vertical as shown in the figure. When the string is cut, which of the illustrated paths best indicates the trajectory of the mass?

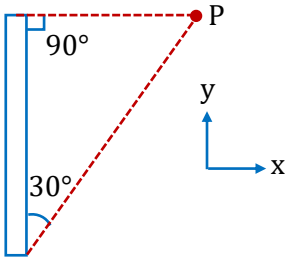


(A) I (B) II (C) III (D) IV

PEO104

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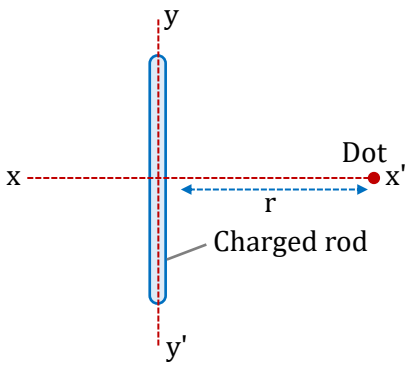
26. The direction (θ) of $\vec{E}$ at point *P* due to uniformly charged finite rod will be :-



(A) At angle 30° from *x*-axis
(B) 45° from *x*-axis
(C) 60° from *x*-axis
(D) None of these

PEO119

27. A uniformly charged rod is kept on *y*-axis with centre at origin, as shown. Which of the following actions will increase the electric field strength at the position of the dot ?

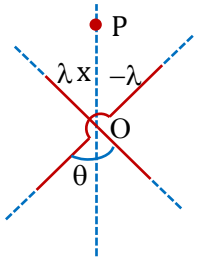


(A) Make the rod longer without changing the charge.
(B) Make the rod shorter without changing the charge.
(C) Make the rod shorter without changing the linear charge density.
(D) Rotate the rod about *yy*'.

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PEO123

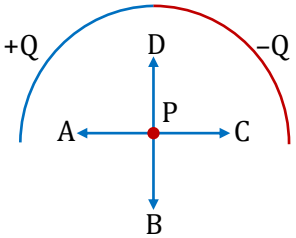
28. If the line charge densities are λ and –λ. as shown in figure. Find the electric field at *P*.



(A) $\frac{2k\lambda}{x}\cot\frac{\theta}{2}$ (B) $\frac{4k\lambda}{x}\cot\frac{\theta}{2}$ (C) $\frac{2k\lambda}{x}\tan\frac{\theta}{2}$ (D) $\frac{4k\lambda}{x}\tan\frac{\theta}{2}$

PEO127

29. As shown in the figure, an insulating rod is set into the shape of a semicircle. The left half of the rod has a charge of + *Q* uniformly distributed along its length, and the right half of the rod has a charge of –*Q* uniformly distributed along its length. Which vector shows the correct direction of the electric field at point *P*, the centre of the semicircle ?



(A) A
(B) B
(C) C
(D) D

PEO120

For More Join: @IITJEE_Advanced

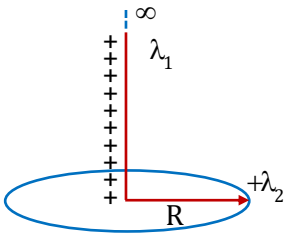
30. A nonconducting ring of radius R has uniformly distributed positive charge Q. A small part of the ring, of length d, is removed ($d << R$). The electric field at the centre of the ring will now be
(A) directed towards the gap, inversely proportional to R^3.
(B) directed towards the gap, inversely proportional to R^2.
(C) directed away from the gap, inversely proportional to R^3.
(D) directed away from the gap, inversely proportional to R^2.

PEO121

31. The maximum electric field at a point on the axis a uniformly charged ring is E_0. At how many points on the axis will the magnitude of electric field be $E_0/2$:-
(A) 1 (B) 2 (C) 3 (D) 4

PEO122

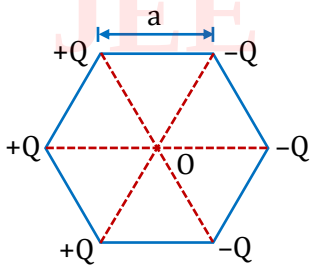
32. A semi-infinite charged wire with charge density $+\lambda_1$ is placed along the axis of a uniformly charged ring of charge density $+\lambda_2$. Find the force exerted on the wire due to the ring. The radius of ring is R.



(A) $(k\lambda_1\lambda_2)\ 2\pi R$ (B) $\frac{\lambda_1\lambda_2}{2\epsilon_0}$ (C) $k\lambda_1^2 2\pi R$ (D) None of these

PEO126

33. Six charges are placed at the vertices of a regular hexagon as shown in the figure. The vertical component of the electric field along the line passing through point O and perpendicular to the plane of the figure at a distance of x (>>a) from O is

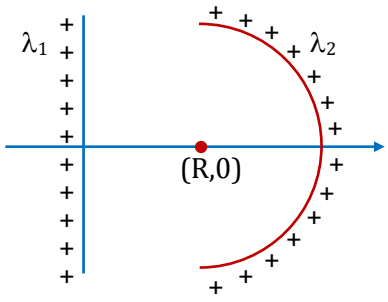


(A) $\frac{Qa}{\pi\varepsilon_0 x^3}$ (B) $\frac{2Qa}{\pi\varepsilon_0 x^3}$

(C) $\frac{\sqrt{3}Qa}{\pi\varepsilon_0 x^3}$ (D) zero

PEO271

34. A uniformly charged infinite wire is placed along 'y' axis having linear charge density 'λ_1'. A semicircle wire having linear charge density 'λ_2' centred at (R, 0) is placed as shown. Find the ratio of $\frac{\lambda_1}{\lambda_2}$. If electric field at ($R$, 0) is zero.



(A) $\frac{\lambda_1}{\lambda_2}=1$ (B) $\frac{\lambda_2}{\lambda_1}=2$

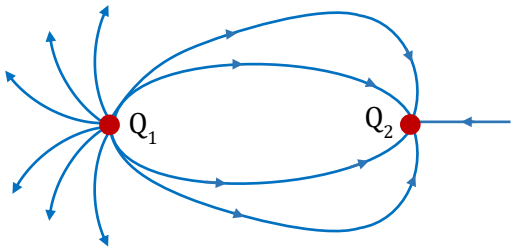
(C) $\frac{\lambda_2}{\lambda_1}=\frac{1}{2}$ (D) $\frac{\lambda_1}{\lambda_2}=\sqrt{2}$

PEO272

For More Join: @IITJEE_Advanced

Properties of Electric Line of Force and Gauss Law

35. From the field pattern shown in figure, identify the correct statement related to the charges Q_1 and Q_2.

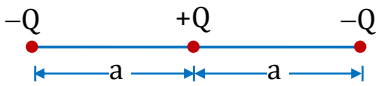


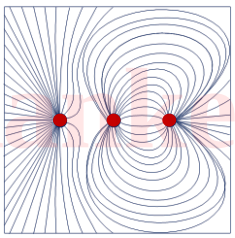
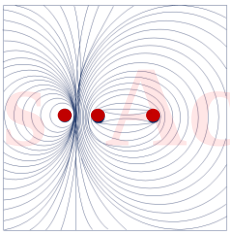
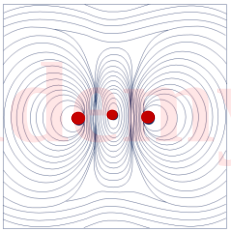
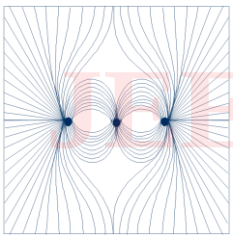
(A) $\frac{Q_1}{Q_2} = +2.5$ (B) $\frac{Q_1}{Q_2} = +2$

(C) $\frac{Q_1}{Q_2} = -2$ (D) There is no relation exist between Q_1 and Q_2

PEO137

36. The fig. shows the distribution of three charges $-Q$, $+Q$ and $-Q$ on the X-axis. Which of the following figures shows the possible electric field lines for the distribution?

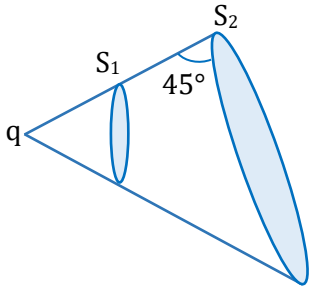


(A)  (B)  (C)  (D) 

Resonance Academy JEE

PEO138

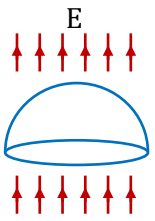
37. In the given figure flux through surface S_1 is f_1 and through S_2 is f_2. Which is correct ?



(A) $f_1 = f_2$ (B) $f_1 > f_2$ (C) $f_1 < f_2$ (D) None of these

PEO135

38. A hemispherical surface (half of a spherical surface) of radius R is located in a uniform electric field E that is parallel to the axis of the hemisphere. What is the magnitude of the electric flux through the hemisphere surface?



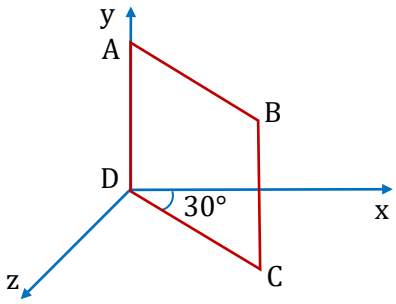
(A) 0 (B) $4\pi R^2E/3$

(C) $2\pi R^2E$ (D) πR^2E

PEO136

For More Join: @IITJEE_Advanced

39. A square cross-section *ABCD* of side a is kept in the direction shown in the diagram. An electric field $E_0\hat{i}+E_0\hat{j}$ exists in the region. The flux of this field through the area cross-section is :



(A) E_0a^2 (B) $\frac{E_0a^2}{3}$ (C) $\frac{E_0a^2}{2}$ (D) $\frac{\sqrt{3}}{2}E_0a^2$

PEO141

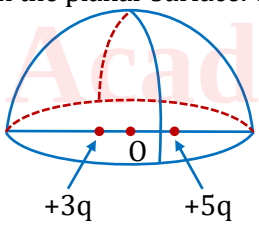
40. An electric field $\vec{E}=E_0\hat{i}+E_0\hat{j}$ exists in a space. Find the flux through a triangular loop with vertices at $\left(\frac{a}{2},0,0\right)$, $\left(\frac{a}{2},0,a\right)$ and $\left(\frac{a}{2},a,\frac{a}{2}\right)$

(A) $\sqrt{2}\,E_0a^2$ (B) $\frac{E_0a^2}{2}$ (C) $\frac{\sqrt{3}}{2}E_0a^2$ (D) $\frac{\sqrt{3}}{4}E_0a^2$

PEO273

41. The figure shows a hollow hemisphere of radius *R* in which two charges $3q$ and $5q$ are placed symmetrically about the centre *O* on the planar surface. The electric flux over the curved surface is:

Rankers Academy JEE



(A) $\frac{15}{2}\frac{q}{\varepsilon_0}$ (B) $\frac{4q}{\varepsilon_0}$ (C) $\frac{q}{\varepsilon_0}$ (D) $\frac{2q}{\varepsilon_0}$

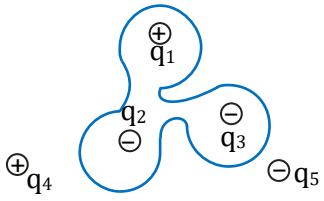
PEO274

42. A very long uniformly charged thread oriented along the axis of a circle of radius $R = 1m$, rests on its centre with one of the ends. The charge per unit length on the thread is $\lambda=16\varepsilon_0$. Find the flux of the vector *E* through the circle area in (*Vm*).

(A) 4 (B) 8 (C) 16 (D) 32

PEO144

43. Figure shows five charged lumps of non-conducting material.
The cross-section of a gaussian surface *S* is indicated. What is the net electric flux through the surface if $q_1 = q_4 = 10\ nC$ $q_2 = q_5 = -10\ nC$ and $q_3 = -5\ nC$



(A) – 565 (B) – 400 (C) – 800 (D) – 500

PEO150

For More Join: @IITJEE_Advanced

44. A square of side 20 *cm* is enclosed by a surface of sphere of 80 *cm* radius. Square and sphere have the same centre. Four charges $+2\times10^{-6}$ *C*, -5×10^{-6} *C*, -3×10^{-6} *C*, $+6\times10^{-6}$ *C* are located at the four corners of a square, then out going total flux from spherical surface in $N-m^2/C$ will be

(A) Zero (B) $(16\pi)\times10^{-6}$ (C) $(8)\times10^{-6}$ (D) 36×10^{-6}

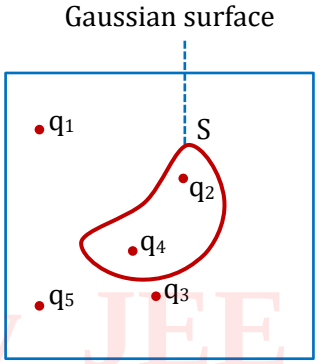
PEO151

45. Gauss's law is given by $\oint_s \vec{E}.d\vec{s} = \frac{q}{\epsilon_0}$, if net charge enclosed by a Gaussian surface is zero then:

(A) *E* must be zero on the surface
(B) Number of incoming and outgoing electric lines are equal
(C) There is a net incoming of electric lines
(D) None

PEO147

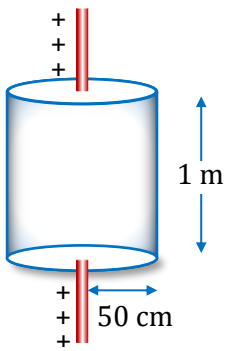
46. Five charges q_1, q_2, q_3, q_4, and q_5 are fixed at their positions as shown in Fig. (*S* is a Gaussian surface). The Gauss's law is given by $\oint E.ds = \frac{q}{\varepsilon_0}$. Which of the following statements is correct?



(A) *E* on the LHS of the above equation will have a contribution from q_1, q_3 and q_5 while *q* on the RHS will have a contribution from q_2 and q_4 only.
(B) *E* on the LHS of the above equation will have a contribution from all charges while *q* on the RHS will have a contribution from q_2 and q_4 only.
(C) *E* on the LHS of the above equation will have a contribution from all charges while *q* on the RHS will have a contribution from q_1, q_3 and q_5 only.
(D) Both *E* on the LHS and *q* on the RHS will have contributions from q_2 and q_4 only.

PEO148

47. Electric charge is uniformly distributed along a long straight wire of radius 1 *mm*. The charge per *cm* length of the wire is *Q* coulomb. Another cylindrical surface of radius 50 *cm* and length 1 *m* symmetrically encloses the wire as shown in the figure. The total electric flux passing through the cylindrical surface is



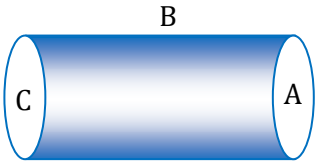
(A) $\frac{Q}{\varepsilon_0}$ (B) $\frac{100Q}{\varepsilon_0}$

(C) $\frac{10Q}{(\pi\varepsilon_0)}$ (D) $\frac{100Q}{(\pi\varepsilon_0)}$

PEO152

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For More Join: @IITJEE_Advanced

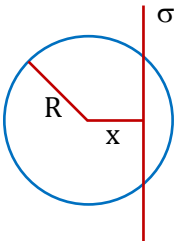
48. A hollow cylinder has a charge q coulomb symmetrically within it. If ϕ is the electric flux in units of Vm associated with the curved surface B, the flux linked with the plane surface A in units of Vm will be-



(A) $\frac{q}{2\varepsilon_0}$ (B) $\frac{\phi}{3}$ (C) $\frac{q}{\varepsilon_0}-\phi$ (D) $\frac{1}{2}\left(\frac{q}{\varepsilon_0}-\phi\right)$

PEO154

49. An infinite uniformly charged sheet with surface charge density σ cuts through a spherical Gaussian surface of radius R at a distance x from its center, as shown in the figure. The electric flux ϕ through the Gaussian surface is :-

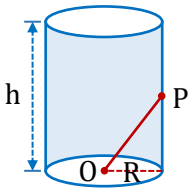


(A) $\frac{\pi R^2\sigma}{\varepsilon_0}$ (B) $\frac{2\pi\left(R^2-x^2\right)\sigma}{\varepsilon_0}$ (C) $\frac{\pi\left(R-x\right)^2\sigma}{\varepsilon_0}$ (D) $\frac{\pi\left(R^2-x^2\right)\sigma}{\varepsilon_0}$

PEO146

Rankers Academy JEE

50. Electric field at point P is given by $\vec{E}=E_0\vec{r}$. The total flux through the given cylinder of radius R and height h is:



(A) $E_0\pi R^2h$ (B) $2E_0\pi R^2h$ (C) $3E_0\pi R^2h$ (D) $4E_0\pi R^2h$

PEO163

51. A solid non conducting sphere having variable volume charge density $d=\frac{Ar}{R}$. If electric field at distance $\frac{R}{2}$ from the centre of charged sphere is $E=\frac{2R}{\in_0}\ N/C$. Value of $\frac{A}{16}$ will be

(A) 2 (B) 4 (C) 6 (D) 8

PEO164

52. Let $P(r)=\frac{Q}{\pi R^4}r$ be the charge density distribution for a solid sphere of radius R and total charge Q. For a point 'p' inside the sphere at distance r_1 from the centre of the sphere, the magnitude of electric field is –

(A) 0 (B) $\frac{Q}{4\pi\in_0 r_1^2}$ (C) $\frac{Qr_1^2}{4\pi\in_0 R^4}$ (D) $\frac{Qr_1^2}{3\pi\in_0 R^4}$

PEO167

For More Join: @IITJEE_Advanced

53. A ball of radius R carries a positive charge whose volume charge density depends only on the distance r form the ball's centre as :

$$\rho = \rho_0\left(1-\frac{r}{R}\right)$$

where ρ_0 is a constant. Assume ε as the permittivity of the ball.

The magnitude of the electric field as a function of the distance r outside the ball is given by :

(A) $E = \frac{\rho_0 R^3}{8\varepsilon r^2}$ (B) $E = \frac{\rho_0 R^3}{12\varepsilon r^2}$ (C) $E = \frac{\rho_0 R^2}{8\varepsilon r^3}$ (D) $E = \frac{\rho_0 R^2}{12\varepsilon r^3}$

PEO168

54. A system consists of uniformly charged sphere of radius R and a surrounding medium filled by a charge with the volume density $\rho = \frac{\alpha}{r}$, where a is a positive constant and r is the distance from the centre of the sphere. The charge of the sphere for which electric field intensity E outside the sphere is independent of r is

(A) $\frac{\alpha}{2\in_0}$ (B) $\frac{2}{\alpha\in_0}$ (C) $2\,paR^2$ (D) aR^2

PEO171

55. The electric field inside a sphere which carries a charge density proportional to the distance from the origin $\rho = \propto r$ ($\propto$ is a constant) is:

(A) $\frac{\propto r^3}{4\varepsilon_0}$ (B) $\frac{\propto r^2}{4\varepsilon_0}$ (C) $\frac{\propto r^2}{3\varepsilon_0}$ (D) None of these

PEO166

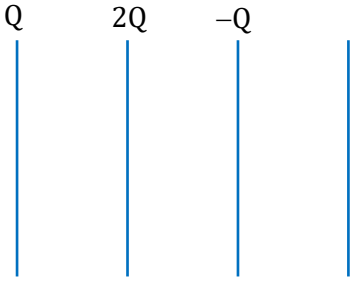
Rankers Academy JEE

56. For a spherically symmetrical charge distribution, electric field at a distance r from the centre of sphere is $\vec{E} = kr^7\hat{r}$. Where k is a constant. What will be the volume charge density at a distance r from the centre of sphere?

(A) $\rho = 9k\varepsilon_0 r^6$ (B) $\rho = 5k\varepsilon_0 r^3$ (C) $\rho = 3k\varepsilon_0 r^4$ (D) $\rho = 9k\varepsilon_0 r^0$

PEO165

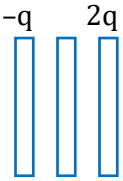
57. Find charge on left surface of the third plate.



(A) Q (B) 2Q (C) –Q (D) –2Q

PEO178

58. Three identical metal plates with large equal surface areas are kept parallel to each other as shown in figure. The leftmost plate is given a charge $-q$, the rightmost a charge $2q$ and the middle one remains neutral. Find the charge appearing on the outer surface of the leftmost plate.

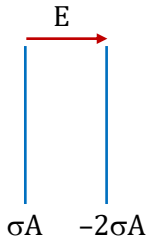


(A) $4q$ (B) $q/4$
(C) $q/2$ (D) $2q$

PEO179

For More Join: @IITJEE_Advanced

59. Two thin conducting plates (very large) parallel to each other carrying total charges σA and $-2\sigma A$ respectively (where A is the area of each plate), are placed in a uniform external electric field E as shown. Find the charge on inner surface of first plate.

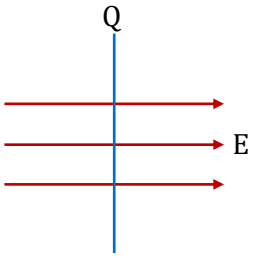


(A) $\left[\frac{3\sigma+2\varepsilon_0 E}{2}\right]A$ (B) $\left[\frac{2\sigma+2\varepsilon_0 E}{3}\right]A$

(C) $\left[\frac{2\sigma+3\varepsilon_0 E}{3}\right]A$ (D) $\left[\frac{2\sigma+3\varepsilon_0 E}{4}\right]A$

PEO180

60. An isolated conducting sheet of area A and carrying a charge Q is placed in a uniform electric field E, such that electric field is perpendicular to sheet and covers all the sheet. Find out charges appearing on right surface of plate.



(A) $\frac{Q}{2} - EA\varepsilon_0$ (B) $\frac{Q}{2} - 2EA\varepsilon_0$

(C) $\frac{Q}{2} + EA\varepsilon_0$ (D) $\frac{Q}{2} + 2EA\varepsilon_0$

PEO181

Electric Potential Energy and Electric Potential

61. Two positively charged particles X and Y are initially far away from each other and at rest. X begins to move towards Y with some initial velocity. The total momentum and energy of the system are p and E.

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(A) If Y is fixed, both p and E are conserved.

(B) If Y is fixed, E is conserved, but not p.

(C) If both are free to move, p is conserved but not E.

(D) If both are free, E is conserved, but not p.

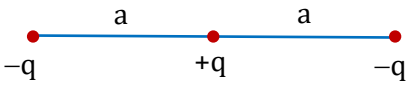
PEO183

62. Potential energy of a system comprising of point charges is U_1. When a charge q is added in the system without disturbing other charges, the potential energy becomes U_2. The potential of the point where the charge q is placed in the system is

(A) $\frac{U_2-U_1}{q}$ (B) $\frac{U_1-U_2}{q}$ (C) $\frac{U_1+U_2}{2q}$ (D) $\frac{U_2-U_1}{2q}$

PEO184

63. Three point charges shown in the figure be along a straight line. The work done against electrostatic forces to exchange the position of the central $+ve$ charge with one of the negative charge is



(A) $\frac{q^2}{4\pi\in_0 a}$ (B) $\frac{q^2}{8\pi\in_0 a}$ (C) $\frac{-q^2}{8\pi\in_0 a}$ (D) $\frac{-q^2}{4\pi\in_0 a}$

PEO187

For More Join: @IITJEE_Advanced

64. A positively charged particle is released from rest in an uniform electric field. The electric potential energy of the charge

(A) remains a constant because the electric field is uniform.

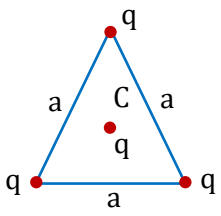
(B) increases because the charge moves along the electric field.

(C) decreases because the charge moves along the electric field.

(D) decreases because the charge moves opposite to the electric field.

PEO186

65. Four equal charges of magnitude q are as shown in fig. Now the charge at center C is taken to infinity slowly, then work done by external force will be -



(A) $\frac{-3Kq^2}{a}$ (B) $\frac{-3\sqrt{3}Kq^2}{a}$

(C) $\frac{-3Kq^2}{a}(\sqrt{3}+1)$ (D) $\frac{3Kq^2}{a}(\sqrt{3}+1)$

PEO188

66. Two identical particles of mass m carry a charge Q each. Initially one is at rest on a smooth horizontal plane and the other is projected along the plane directly towards first particle from a large distance with speed v. The closest distance of approach will be

(A) $\frac{1}{4\pi\varepsilon_0}\frac{Q^2}{m\nu}$ (B) $\frac{1}{4\pi\varepsilon_0}\frac{4Q^2}{m\nu^2}$ (C) $\frac{1}{4\pi\varepsilon_0}\frac{2Q^2}{m\nu^2}$ (D) $\frac{1}{4\pi\varepsilon_0}\frac{3Q^2}{m\nu^2}$

PEO189

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67. A charge + q is placed at each of the points $x = x_0, x = 3x_0, x = 5x_0$,...., and infinitum on the x-axis, and a charge – q is placed at each of the points $x = 2x_0, x = 4x_0, x = 6x_0$ .... and infinitum. Here x_0 is a positive constant. Take the electric potential at a point due to a charge q at a distance r from it to be $Q/(4\pi\varepsilon_0 r)$. Then the potential at the origin due to the above system of charge is -

(A) 0 (B) $\frac{q}{8\pi\varepsilon_0 \log_e 2}$ (C) ∞ (D) $\frac{q\log_e 2}{4\pi\varepsilon_0 x_0}$

PEO196

68. Eight charges, each of the magnitude q are placed at the vertices of a cube placed in vacuum. Electric potential at the centre of the cube due to this system of charges is -

(ε_0 is permittivity of vacuum and a is length of the each side of the cube)

(A) $\frac{2q}{\pi\varepsilon_0 a}$ (B) $\frac{4q}{\sqrt{3}\pi\varepsilon_0 a}$ (C) zero (D) $\frac{\sqrt{3}q}{\pi\varepsilon_0 a}$

PEO194

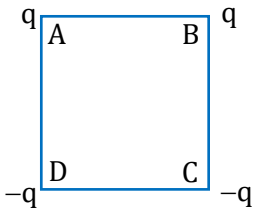
69. Four charges of $6\mu C$, $2\mu C$, $-12\mu C$ and $4\mu C$ are placed at the corners of a square of side $1m$. The square is in x-y plane and its center at its origin. Electric potential due to these charges is zero everywhere on the line -

(A) $x = y, z = 0$ (B) $x = 0 = z$ (C) $x = 0 = y$ (D) None of these

PEO193

For More Join: @IITJEE_Advanced

70. Charges are placed on the vertices of a square as shown. Let $\vec{E}$ be the electric field and V the potential at the centre. If the charges on A and B are interchanged with those on D and C respectively, then –



(A) $\vec{E}$ remains unchanged, V changes

(B) Both $\vec{E}$ and V changes

(C) $\vec{E}$ and V remains unchanged

(D) $\vec{E}$ changes, V remains unchanged

PEO192

71. Two fixed charges A and B of 5 μC each are separated by a distance of 6 m. C is the mid-point of the line joining A and B. A charge 'Q' of $-5\mu C$ is shot perpendicular to the line joining A and B through C with a kinetic energy of $0.06J$. The charge 'Q' comes to rest at a point D. The distance CD is:-

(A) 3 m (B) $\sqrt{3}$ m (C) $3\sqrt{3}$ m (D) 4 m

PEO221

72. At any point $(x,0,0)$ the electric potential V is $\left(\frac{1000}{x}+\frac{1500}{x^2}+\frac{500}{x^3}\right)$ volt, then electric field at $x = 1$ m will be :-

(A) $5500\left(\hat{j}+\hat{k}\right)V/m$ (B) $5500\hat{i}V/m$ (C) $\frac{5500}{\sqrt{2}}\left(\hat{j}+\hat{k}\right)V/m$ (D) $\frac{5500}{\sqrt{2}}\left(\hat{i}+\hat{k}\right)V/m$

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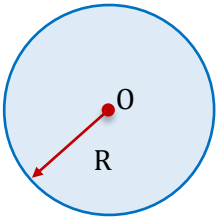
PEO199

73. In an electric field the potential at a point is given by the following relation $V=\frac{343}{r}$. The electric field at $\vec{r}=3\hat{i}+2\hat{j}+6\hat{k}$ is :

(A) $21\hat{i}+14\hat{j}+42\hat{k}$ (B) $3\hat{i}+2\hat{j}+6\hat{k}$ (C) $\frac{1}{7}\left(3\hat{i}+2\hat{j}+6\hat{k}\right)$ (D) $-\left(3\hat{i}+2\hat{j}+6\hat{k}\right)$

PEO200

74. Volume charge density of a non–conducting sphere is varying as $\rho = \rho_0 r$ where ρ_0 is a constant and r is the distance from centre of sphere. If potential at $r = 2R$ is zero then potential at infinity will be

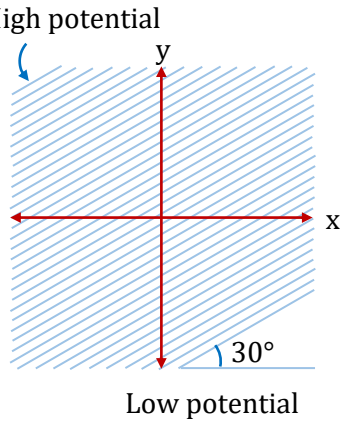


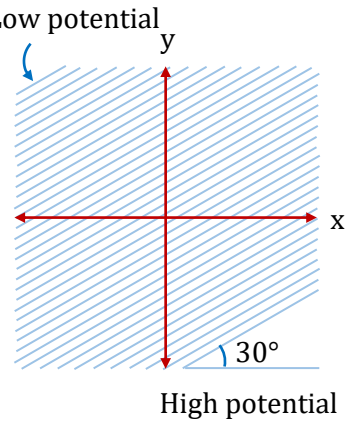
(A) $\frac{\rho_0 R^3}{8\epsilon_0}$ (B) $-\frac{\rho_0 R^3}{8\epsilon_0}$ (C) $-\frac{\rho_0 R^3}{4\epsilon_0}$ (D) Data insufficient

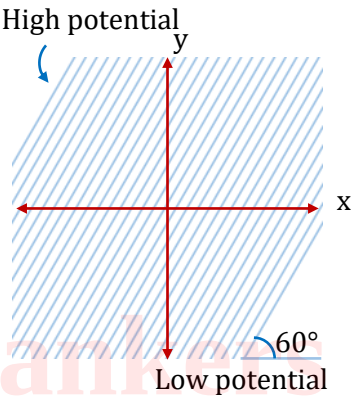
PEO212

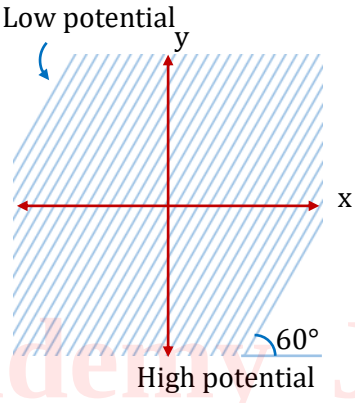
For More Join: @IITJEE_Advanced

75. The electric field intensity at all points in space is given by $\vec{E} = \sqrt{3}\hat{i} - \hat{j}$ volts/metre. The nature of equipotential lines in *x*-*y* plane is given by :-

(A)


(B)


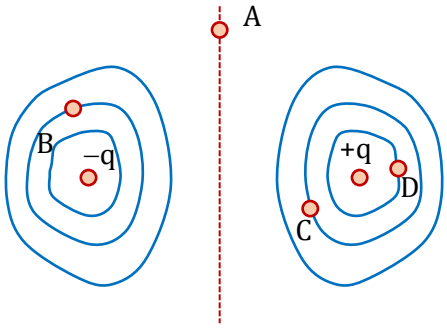
(C)


(D)


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PEO213

76. Figure shows equi-potential surfaces for a two charges system. At which of the labeled points point will an electron have the highest potential energy?



(A) Point A (B) Point B (C) Point C (D) Point D

PEO191

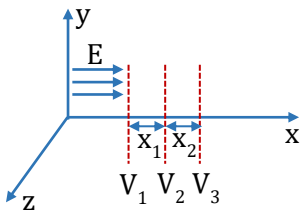
77. Equipotential at a great distance from a collection of charges whose total sum is not zero are approximately

(A) spheres (B) planes (C) paraboloids (D) ellipsoids

PEO214

For More Join: @IITJEE_Advanced

78. In an electric field shown in figure three equipotential surfaces are shown. If function of electric field is $E = 2x^2V/m$, and given that $V_1 - V_2 = V_2 - V_3$, then we have



(A) $x_1 = x_2$
(B) $x_1 > x_2$
(C) $x_2 > x_1$
(D) Data insufficient

PEO215

79. Consider the following conclusions regarding the components of an electric field at a certain point in space given by

$E_x = -Ky$ $E_y = Kx$ $E_z = 0$

(I) The field is conservative. (II) The field is non-conservative.
(III) The lines of force are straight lines (IV) The lines of force are circles.
Of these conclusions
(A) II and IV are valid (B) I and III are valid
(C) I and IV are valid (D) II and III are valid

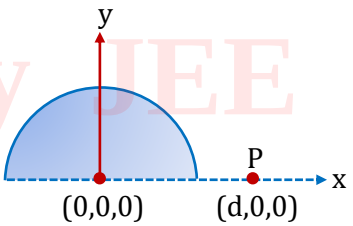
PEO216

80. Magnitude of electric field depends only on the x-coordinate given by $\vec{E} = \frac{20}{x^2}\hat{i}$ V/m. Find the potential difference between two points A $(5m, 0)$ and B $(10m, 0)$.

(A) -5 V (B) 5 V (C) -2 V (D) 5 V

PEO224

81. A solid hemispherical uniform charged body having charge Q is kept symmetrically along the y-axis as shown in figure. The electric potential at a distance d from the origin along the x-axis at point P will be



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(A) $\frac{1}{4\pi\varepsilon_0}\frac{Q}{d}$

(B) less than $\frac{1}{4\pi\varepsilon_0}\frac{Q}{d}$

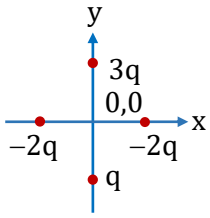
(C) more than $\frac{1}{4\pi\varepsilon_0}\frac{Q}{d}$ and less then $\frac{2}{4\pi\varepsilon_0}\frac{Q}{d}$

(D) more than $\frac{2}{4\pi\varepsilon_0}\frac{Q}{d}$

PEO275

Dipole

82. 4 charges are placed each at a distance 'a' from origin. The dipole moment of configuration is-

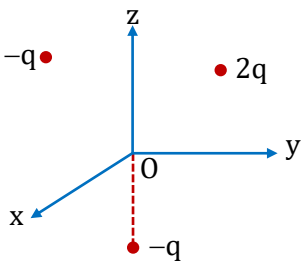


(A) $2qa\hat{j}$ (B) $3qa\hat{j}$ (C) $2aq\left[\hat{i}+\hat{j}\right]$ (D) None of these

PEO230

For More Join: @IITJEE_Advanced

83. Three point charges $2q$, $-q$ and $-q$ are located respectively at $(0, a, a)$, $(0, a, -a)$ and $(0, 0, -a)$ as shown. The dipole moment of this distribution is :-



(A) $2qa$ in the y–z plane at $\tan^{-1}\left(\frac{1}{4}\right)$ with z-axis

(B) $\sqrt{17}qa$ in the y–z plane at $\tan^{-1}\left(\frac{1}{4}\right)$ with z-axis

(C) $\sqrt{5}qa$ in the x–y plane at $\tan^{-1}(4)$ with y-axis

(D) $4qa$ in the x–y plane at $\tan^{-1}(4)$ with y-axis

PEO231

84. Three point charges $+q$, $-2q$ and $+q$ are placed at points $(x = 0, y = a, z = 0)$, $(x = 0, y = 0, z = 0)$ and $(x = a, y = 0, z = 0)$ respectively. The direction of the electric dipole moment vector of this charge assembly is :–

(A) along + x direction

(B) along + y direction

(C) along the line joining points $(x = 0, y = 0, z = 0)$ and $(x = a, y = a, z = 0)$

(D) along the line joining points $(x = 0, y = 0, z = 0)$ and $(x = a, y = 0, z = 0)$

PEO229

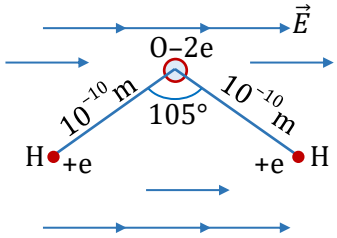
85. An electric dipole of dipole moment $\vec{p}$ is placed at the origin along the x-axis. The angle made by electric field with x-axis at a point P, whose position vector makes an angle θ with x-axis, is :

(Where, $\tan\alpha = \frac{1}{2}\tan\theta$)

(A) α (B) θ (C) $\theta + \alpha$ (D) $\theta + 2\alpha$

PEO234

86. A water molecule as shown is in a region of uniform electric field $\vec{E} = 1000\hat{i}\ V/m$. This molecule experiences



(A) A counter clockwise torque

(B) A clockwise torque

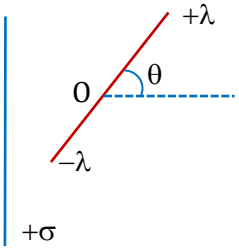
(C) A net force to the right

(D) A net force to the left

PEO242

For More Join: @IITJEE_Advanced

87. A large sheet carries uniform surface charge density *s*. A rod of length 2ℓ has a linear charge density λ on one half and $-\lambda$ on the other half. The rod is hinged at mid point *O* and makes angle θ with the normal to the sheet. The torque experienced by the rod is :-



(A) $\frac{\sigma\lambda\ell^2}{2\varepsilon_0}\cos\theta$ (B) $\frac{\sigma\lambda\ell}{\varepsilon_0}\cos 2\theta$ (C) $\frac{\sigma\lambda\ell^2\sin\theta}{2\varepsilon_0}$ (D) $\frac{\sigma\lambda\ell\sin^2\theta}{\varepsilon_0}$

PEO244

88. An electric dipole (dipole moment *p*) is placed at a radial distance $r >> a$ (where a is dipole length) from a infinite line of charge having linear charge density $+\lambda$. Dipole moment vector is aligned along radial vector $\vec{r}$ force experienced by dipole is :-

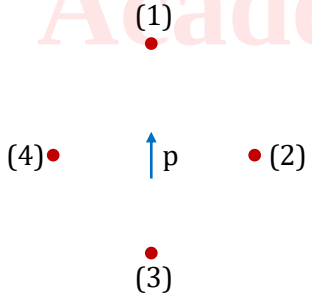
(A) $\frac{\lambda p}{2\pi\varepsilon_0 r^2}$, attractive (B) $\frac{\lambda p}{2\pi\varepsilon_0 r^3}$, attractive

(C) $\frac{\lambda p}{2\pi\varepsilon_0 r^2}$, repulsive (D) $\frac{\lambda p}{2\pi\varepsilon_0 r^3}$, repulsive

PEO245

89. The drawing shows four points surrounding an electric dipole. Which one of the following expressions best ranks the electric potential at these four locations?

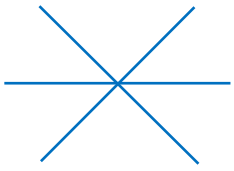
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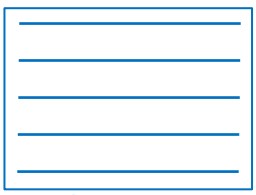


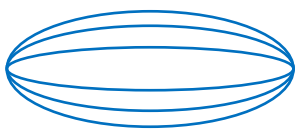
(A) 1 > 2 = 4 > 3 (B) 3 > 2 > 4 > 1 (C) 3 > 2 = 4 > 1 (D) 2 = 4 > 1 = 3

PEO227

90. Which of the following represents the equipotential lines of a dipole (two equal and opposite charges placed at small separation)?

(A) 

(B) 

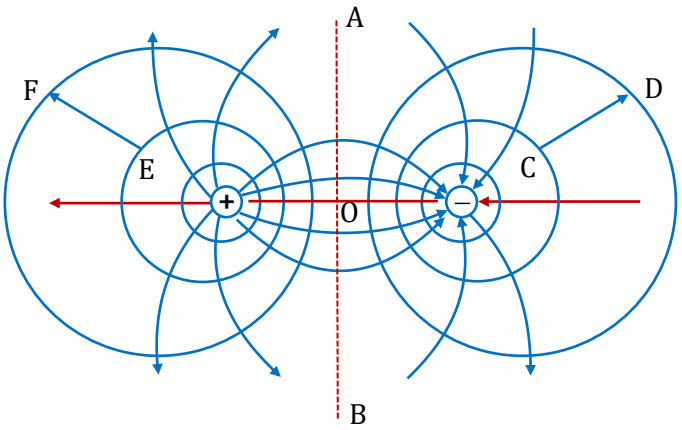
(C) 

(D) 

PEO246

For More Join: @IITJEE_Advanced

91. Figure shows a charge configuration with its equipotential surfaces and electric field lines placed in x-y plane. Mark **INCORRECT** statement :-



(A) Potential gradient is positive along $\left(\overrightarrow{CD}\right)$

(B) Potential gradient is negative along $\left(\overrightarrow{EF}\right)$

(C) If a positive charge particle is constrained to move along line AB, it will have constant potential energy.

(D) A positive charge particle released from rest at C will begin to move in direction $\left(\overrightarrow{DC}\right)$

PEO247

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Conductors

92. The electrostatic potential on the surface of a charged conducting sphere is $100V$. Two statements are made in this regard:

S_1 : At any point inside the sphere, electric field intensity is zero.

S_2 : At any point inside the sphere, the electrostatic potential is $100V$.

Which of the following is a correct statement?

(A) S_1 is true but S_2 is false.

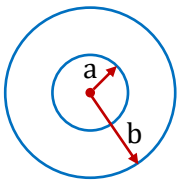
(B) Both S_1 & S_2 are false.

(C) S_1 is true, S_2 is also true and S_1 is the cause of S_2.

(D) S_1 is true, S_2 is also true but the statements are independent.

PEO249

93. If the electric potential of the inner metal sphere is 10 volt & that of the outer shell is 5 volt, then the potential at the centre will be :



(A) 10 volt (B) 5 volt (C) 15 volt (D) 0

PEO251

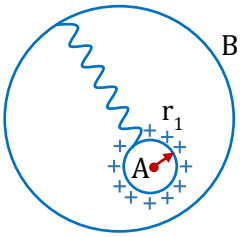
For More Join: @IITJEE_Advanced

94. There are two uncharged identical metallic spheres 1 and 2 of radius r separated by a distance d ($d >> r$). A charged metallic sphere of same radius having charge q is touched with one of the sphere. After some time it is moved away from the system. Now the uncharged sphere is earthed. Charge on earthed sphere is

(A) q (B) $-q$ (C) $-qr/2d$ (D) 0

PEO252

95. A metal sphere A of radius r_1 charged to a potential f_1 is enveloped by a thin walled conducting spherical shell B of radius r_2. Then f_2 of the sphere A after it is connected by a thin wire to the shell B will be :-

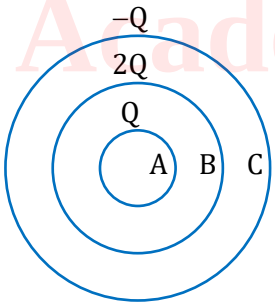


(A) $f_1\frac{r_1}{r_2}$ (B) $f_1\left(\frac{r_2}{r_1}\right)$ (C) $f_1\left(1-\frac{r_1}{r_2}\right)$ (D) $f_1\left(\frac{r_1r_2}{r_1+r_2}\right)$

PEO253

96. Charge Q, $2Q$ and $-Q$ are given to three concentric conducting spherical shells A, B and C respectively. The ratio of charges on the inner and the outer surfaces of the shell 'C' will be :-

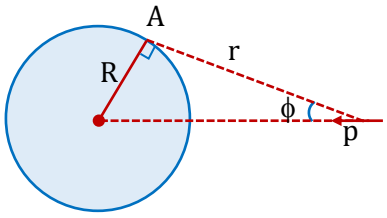
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(A) $\frac{3}{4}$ (B) $-\frac{3}{4}$ (C) $\frac{3}{2}$ (D) $-\frac{3}{2}$

PEO250

97. A dipole having dipole moment p is placed in front of a solid uncharged conducting sphere as shown in the diagram. The net potential at point A lying on the surface of the sphere is :-

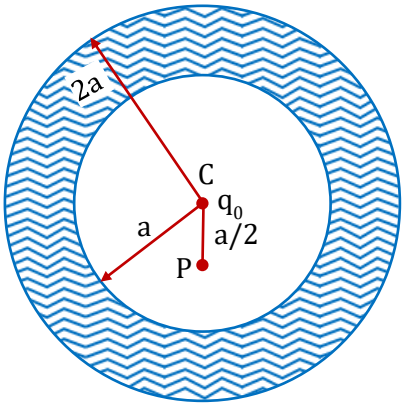


(A) $\frac{kp\cos\phi}{r^2}$ (B) $\frac{kp\cos^2\phi}{r^2}$ (C) 0 (D) $\frac{2kp\cos^2\phi}{r^2}$

PEO254

For More Join: @IITJEE_Advanced

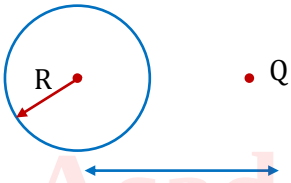
98. A point positive charge q_0 is placed at a point C inside the cavity in a neutral spherical conductor as shown in figure. Find the potential at P :-



(A) $\frac{Kq_0}{2a}$ (B) $\frac{Kq_0}{a}$ (C) $\frac{2Kq_0}{a}$ (D) $\frac{3Kq_0}{2a}$

PEO257

99. A point charge Q is placed near a conducting neutral sphere of radius R. Then electric potential due to induce charges, on the sphere, at the centre of the sphere is
(Given: 'r' is the distance between Centre of sphere and point charge Q).

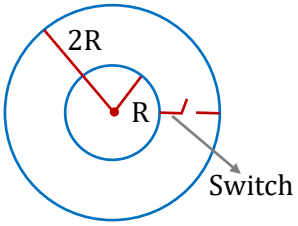


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(A) $\frac{KQ}{r}$ (B) $KQ\left(\frac{1}{R}-\frac{1}{r}\right)$ (C) Zero (D) $\frac{KQ}{R}$

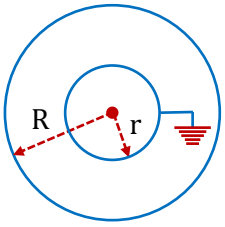
PEO277

100. Net positive charge on inner spherical conductor is $3Q$ and on outer spherical conductor is Q. Switch is closed.
(A) Charge will flow from inner shell to outer shell
(B) Charge will flow from outer shell to inner shell
(C) Charge will not flow
(D) none of these



PEO259

101. If bigger hollow sphere has charge Q then the charge on inner earthed sphere, is :-

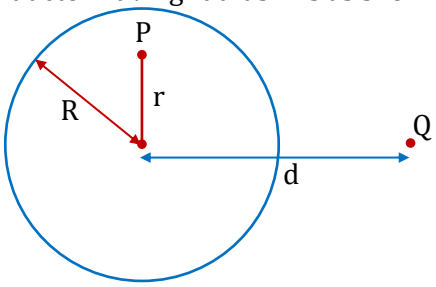


(A) $\frac{R}{r}Q$ (B) $-\frac{R}{r}Q$ (C) $\frac{r}{R}Q$ (D) $-\frac{r}{R}Q$

PEO258

For More Join: @IITJEE_Advanced

102. An uncharged spherical conductor having radius R is as shown. Find potential at a point P :-

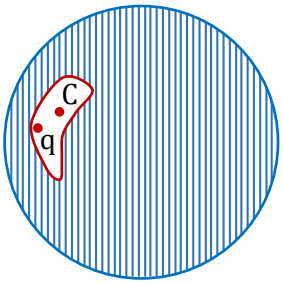


(A) $\frac{kQ}{d}$ (B) $\frac{kQ}{r}$ (C) $\frac{kQ}{R}$ (D) $\frac{kQ}{\sqrt{r^2+d^2}}$

PEO265

103. The figure shows a charge q placed inside a cavity in an uncharged conductor. Now if an external electric field is switched on then :

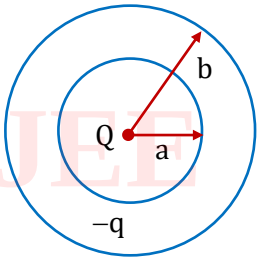
(A) only induced charge on outer surface will redistribute
(B) only induced charge on inner surface will redistribute
(C) Both induced charge on outer and inner surface will redistribute
(D) force on charge q placed inside the cavity will change



PEO266

104. A spherical shell with an inner radius 'a' and an outer radius 'b' is made of conducting material. A point charge $+Q$ is placed at the centre of the spherical shell and a total charge $-q$ is placed on the shell. Charge $-q$ is distributed on the surfaces as

(A) $-Q$ on the inner surface, $-q$ on outer surface
(B) $-Q$ on the inner surface, $-q$ $+Q$ on the outer surface
(C) $+Q$ on the inner surface, $-q$ $-Q$ on the outer surface
(D) The charge $-q$ is spread uniformly between the inner and outer surface.



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PEO248

105. The intensity of an electric field depends only on the coordinates x, y and z as follows :

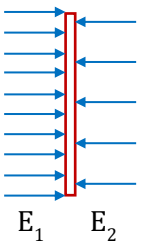
$$\vec{E} = a\frac{\left(x\hat{i}+y\hat{j}+z\hat{k}\right)}{\left(x^2+y^2+z^2\right)^{3/2}}\text{ unit}$$

The electrostatic energy stored between two imaginary concentric spherical shells of radii R and $2R$ with centre at origin is :-

(A) $\frac{4\pi\varepsilon_0 a^2}{R}$ (B) $\frac{2\pi\varepsilon_0 a^2}{R}$ (C) $\frac{\pi\varepsilon_0 a^2}{R}$ (D) $\frac{\pi\varepsilon_0 a^2}{2R}$

PEO255

106. A charged large metal sheet is placed into uniform electric field, perpendicularly to the electric field lines. After placing the sheet into the field, the electric field on the left side of the sheet is $E_1 = 5\times10^5\ V/m$ and on the right it is $E_2 = 3\times10^5\ V/m$. The sheet experiences a net electric force of 0.08 N. Find the area of one face of the sheet. Assume external field to remain constant after introducing the large sheet. Use $\left(\frac{1}{4\pi\varepsilon_0}\right) = 9\times10^9\ Nm^2/C^2$



(A) $3.6\ \pi\times10^{-2}\ m^2$ (B) $0.9\ \pi\times10^{-2}\ m^2$ (C) $1.8\ \pi\times10^{-2}\ m^2$ (D) none

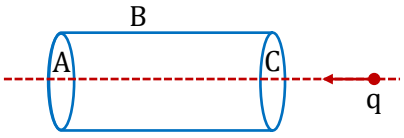
PEO256

For More Join: @IITJEE_Advanced

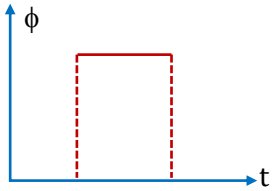
EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. Consider a Gaussian surface as shown. *A* and *C* are the circular faces while *B* is the curved surface. A charge *q* is moved along the axis of cylinder with constant velocity *v*. Choose the correct variation of flux through the surface / cylinder.

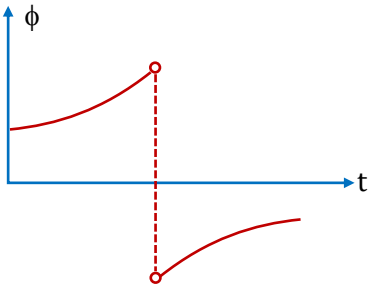


(A)



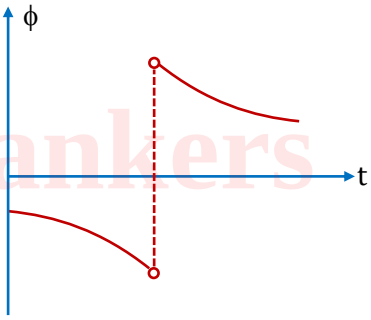
Flux through the cylinder

(B)



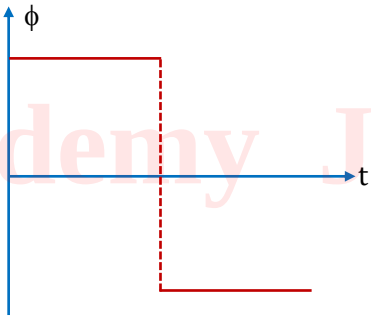
Through surface A

(C)



Through surface C

(D)

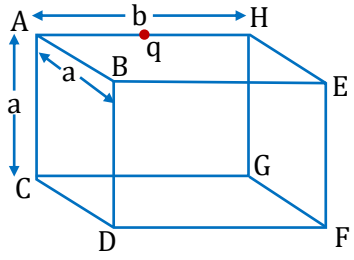


Through surface B

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PEO278

2. A cuboid of dimension [$a \times a \times b$]. Charge *q* is placed at the centre of edge having length '*b*'. If flux through face 'ABCD' is $\frac{q}{32\epsilon_0}$ then select the correct statement(s) :-



(A) Flux through the entire cuboid is $\frac{q}{4\epsilon_0}$

(B) Flux through the face 'ABEH is zero.

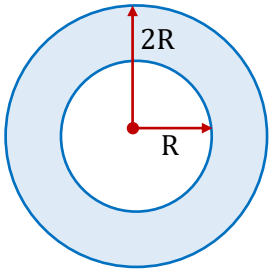
(C) Flux through the face 'BEFD' is $\frac{3q}{32\epsilon_0}$

(D) Flux through the face 'BEFD' is $\frac{q}{16\epsilon_0}$

PEO279

For More Join: @IITJEE_Advanced

3. Consider a spherical volume which has uniform charge density ρ from $x = R$ to $x = 2R$. There is no charge from $x = 0$ to $x = R$ and from $x = 2R$ to infinity. The electric potential due to this charge.



(A) at $x = 0$ is 0

(B) at $x = \frac{R}{2}$ is $\frac{3\rho R^2}{2\epsilon_0}$

(C) at $x = 2R$ is $\frac{7\rho R^2}{6\epsilon_0}$

(D) at $x = \frac{3R}{2}$ is $\frac{8\rho R^2}{3\epsilon_0}$

PEO210

4. The electric field in x-y plane is given by $\vec{E} = x^2\hat{i} + y^2\hat{j}$. Potential at point (3, 3) is taken as zero. Then :-

(A) Work done in moving a test charge q slowly from a point (a, b) to a point (b, a) is zero.

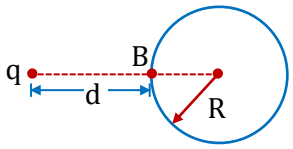
(B) Potential at any point on x-y plane can't exceed 18 unit for x and y in 1st quadrant.

(C) Potential can't be defined as field is non- conservative in nature.

(D) Equipotential curve are concentric circles with centre (3, 3).

PEO280

5. For the situation shown in the figure below, mark out the correct statement(s)



Hollow neutral conductor

(A) Potential of the conductor is $\frac{q}{4\pi\varepsilon_0(d+R)}$

(B) Potential of the conductor is $\frac{q}{4\pi\varepsilon_0 d}$

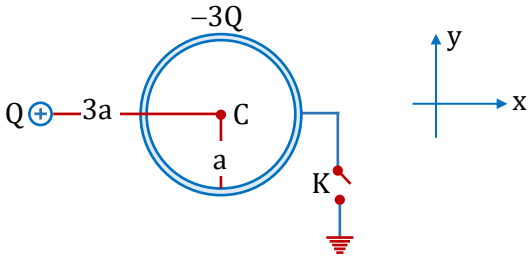
(C) Potential of the conductor can't be determined as nature of distribution of induced charges is not known

(D) Potential at point B due to induced charges is $\frac{-qR}{4\pi\varepsilon_0(d+R)d}$

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PEO261

6. A point charge $+Q$ is kept at a distance $3a$ from the centre of a spherical conducting shell of charge $-3Q$ and radius a. Assuming $K = \frac{1}{4\pi\varepsilon_0}$, mark the **CORRECT** statement(s) :-



(A) Before closing switch, the electric field at C due to the spherical shell is $\left(-\frac{KQ}{9a^2}\hat{i}\right)$

(B) Before closing switch, the potential of the shell is $\left(-\frac{8KQ}{3a}\right)$

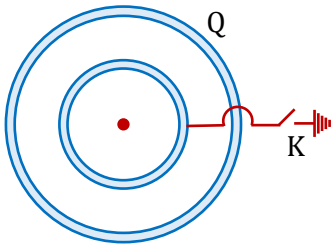
(C) After closing the key, the charge on the shell is $\left(-\frac{Q}{3}\right)$

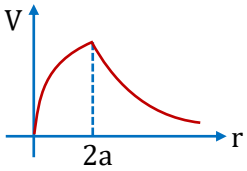
(D) After closing the key, the charge of the shell is zero

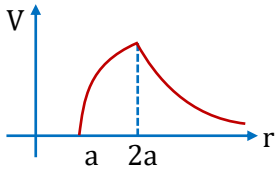
PEO262

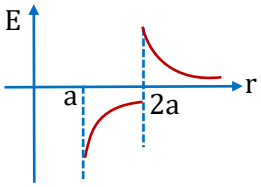
For More Join: @IITJEE_Advanced

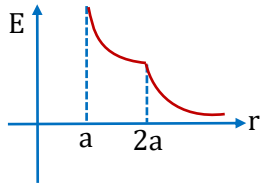
7. In the given figure two concentric conducting spherical shells of radii a and $2a$ have charges zero and Q respectively. When the key K is closed :-



(A) 

(B) 

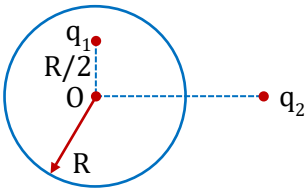
(C) 

(D) 

PEO263

8. As shown in figure a point charge q_1 is placed inside a neutral conducting shell at a distance $\frac{R}{2}$ from the centre of shell. Another point charge q_2 is placed at a distance $\frac{\sqrt{7}}{2}R$ from the centre of shell. Choose the correct option(s).

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(A) The magnitude of force on q_1 due to induced charges at the inner surface of the conducting shell is zero.

(B) The magnitude of force on q_1 due to induced charges at the outer surface of the conducting shell is $\frac{q_1 q_2}{8\pi\varepsilon_0 R^2}$

(C) The magnitude of force on q_2 due to induced charges at the inner surface of the shell is zero.

(D) The magnitude of force on q_2 due to induced charges at the inner surface of the shell is $\frac{q_1 q_2}{8\pi\varepsilon_0 R^2}$

PEO264

For More Join: @IITJEE_Advanced

COMPREHENSION TYPE QUESTIONS

PARAGRAPH FOR QUESTION NO. 9 AND 10

A long cylinder of radius a is charged with a volume charge density

$\rho = \rho_0 r^3$ for $r \le a$

$= 0$ for $r > a$

Where ρ_0 is constant and r is the distance measured from the axis of the cylinder.

9. The charge contained in ℓ length of the cylinder for $r > a$ is :

(A) $\frac{2\pi\rho_0 a^3 \ell}{3}$ (B) $\frac{2\pi\rho_0 a^5 \ell}{5}$ (C) $\frac{4\pi\rho_0 a^3 \ell}{3}$ (D) zero

PEO169

10. The electric field at $r < a$.

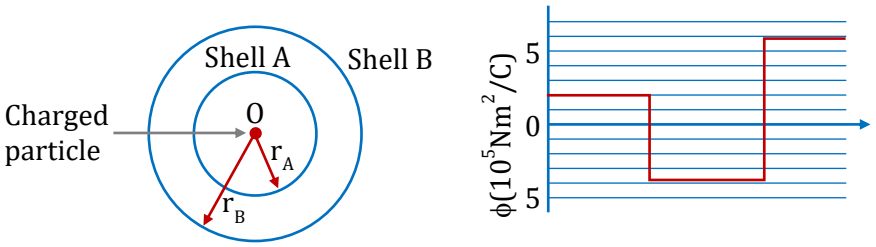
(A) $\frac{\rho_0 r^4}{5\epsilon_0}$ (B) $\frac{\rho_0 r^3}{3\epsilon_0}$ (C) $\frac{\rho_0 r^5}{5\epsilon_0}$ (D) zero

PEO170

PARAGRAPH FOR QUESTION NO. 11 TO 13

A charged particle is suspended at the centre of two thin concentric spherical charged shells, made of non conducting material. Figure *A* shows cross section of the arrangement. Figure *B* gives the net flux ϕ through a Gaussian sphere centered on the particle, as a function of the radius r of the sphere.

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11. What is the charge on the central particle?

(A) 0.2 μC (B) 2μC (C) 1.77 μC (D) 3.4 μC

PEO172

12. What is the charge on shell *A*?

(A) $5.31 \times 10^{-6} C$ (B) $-5.31 \times 10^{-6} C$ (C) $-3.54 \times 10^{-6} C$ (D) $-1.77 \times 10^{-6} C$

PEO173

13. In which range of the values of r is the electric field zero?

(A) 0 to r_A (B) r_A to r_B

(C) for $r > r_B$ (D) for no range of r, electric field is zero.

PEO174

For More Join: @IITJEE_Advanced

MATRIX MATCH TYPE QUESTION

14. In the following question List-I shows various electric field and List-II shows its plot in *XY* plane. Match the Lists appropriately.

List-I

(P) $\vec{E} = y\hat{i} - x\hat{j}$

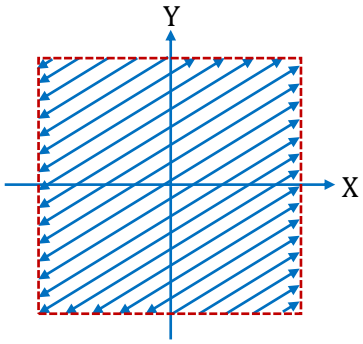
(Q) $\vec{E} = y\hat{i} + y\hat{j}$

(R) $\vec{E} = x\hat{i} + x\hat{j}$

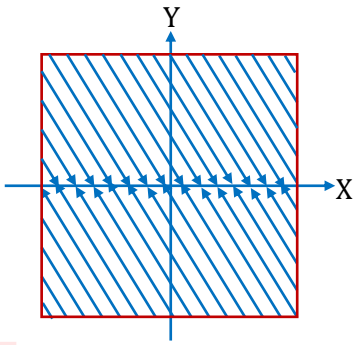
(S) $\vec{E} = y\hat{i} - y\hat{j}$

List-II

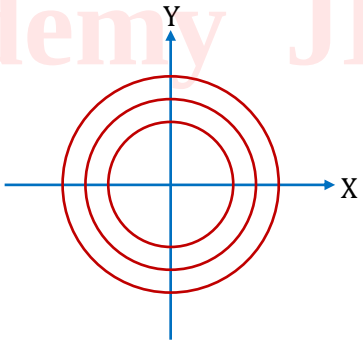
(1)



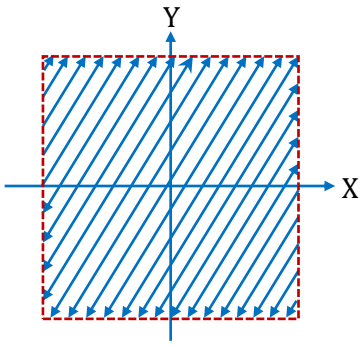
(2)



(3)



(4)



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Code :-

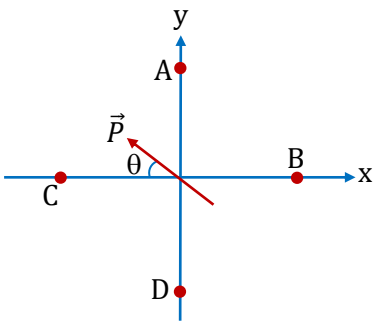
	P	Q	R	S
(A)	3	4	1	2
(C)	2	1	4	3

	P	Q	R	S
(B)	4	2	1	3
(D)	1	2	3	4

PEO281

For More Join: @IITJEE_Advanced

15. Electric dipole is placed at origin as shown. Point *A, B, C* and *D* are at distance *r* from origin, then match the List-I with List-II.



List-I		**List-II**	
(P)	Electric potential at *A*	(1)	$\frac{-kP\sin\theta}{r^2}$
(Q)	Electric potential at *B*	(2)	$\frac{kP\cos\theta}{r^2}$
(R)	Electric potential at *C*	(3)	$\frac{kP\sin\theta}{r^2}$
(S)	Electric potential at *D*	(4)	$\frac{-kP\cos\theta}{r^2}$

Code:

	P	**Q**	**R**	**S**		**P**	**Q**	**R**	**S**
(A)	3	4	2	1	(B)	4	3	1	2
(C)	1	2	4	3	(D)	3	4	1	2

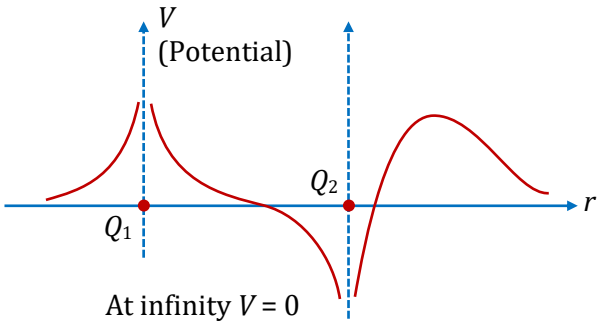
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PEO236

16. As shown in column I their are graphs of electric field (*E*) and potential (*V*) along the line joining charges Q_1 and Q_2 are drawn against distance (*r*) on *x*-axis for charges Q_1 and Q_2. Take potential at infinity equal to zero. [Take direction of *E* in rightward direction as positive]

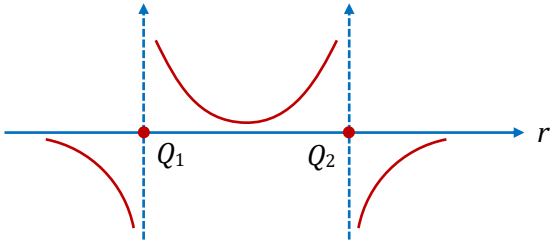
Column-I **Column-II**

(A)



(P) $Q_1 < 0, Q_2 > 0$

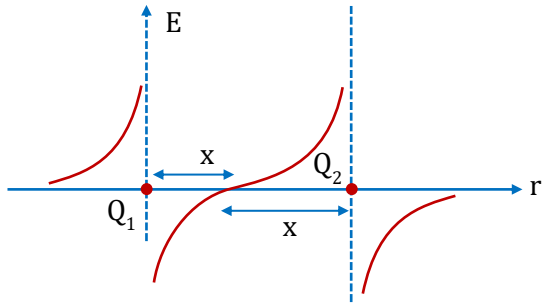
(B)



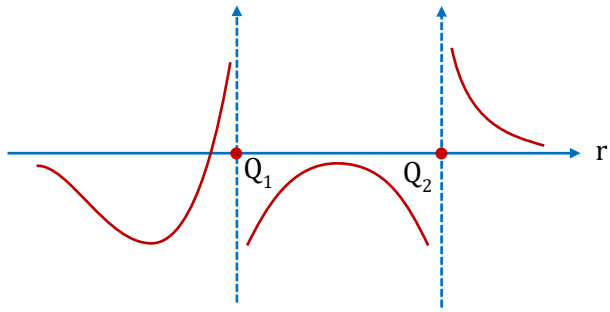
(Q) $Q_1 > 0, Q_2 < 0$

For More Join: @IITJEE_Advanced

(C)



(D)



(R) $|Q_1| > |Q_2|$

(S) $Q_1 < 0, Q_2 < 0$

(T) $|Q_1| = |Q_2|$

PEO197

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For More Join: @IITJEE_Advanced

EXERCISE (S)

Properties of Charge and Coulomb's Law

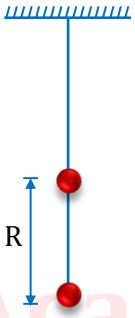
1. What is the excess number of electrons that must be placed on each of two small spheres spaced 3 *cm* apart so that a force of repulsion that acts between them becomes 10^{-19} *N*?

PEO080

2. If charge on each of two point charges is tripled and separation between them is doubled, by what factor will the mutual force between them change?

PEO081

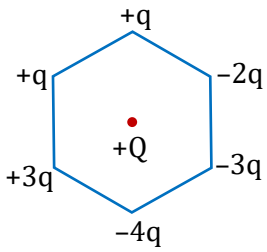
3. Two identical balls of mass $m = 0.9\ g$ each are charged by the same charges, joined by a thread and suspended from the ceiling (figure). Find the charge (in μC) that each ball should have so that the tension in both the threads are same? The distance between the centers of balls is $R = 3m$.



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PEO087

4. Six charges are kept at the vertices of a regular hexagon as shown in the figure. If magnitude of force applied by $+Q$ on $+q$ charge is F, then net electric force on the $+Q$ is nF. Find the value of n.



PEO088

5. Two charged balls are connected by an inextensible thread of length 3 *m*. Masses of balls are 2*kg* and 1 *kg*, the charges are +20 μC and −100 μC. What minimum constant external force F (in N) must be applied to the ball of mass 1 *kg* so that the thread does not slack? Neglect gravity and friction.

PEO089

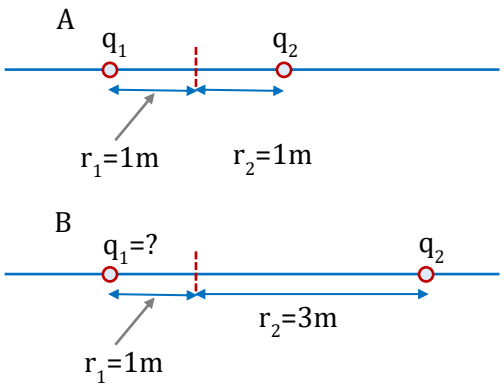
6. A thin conducting ring of radius r has an electric charge $+Q$. If a point charge q is placed at the center of the ring, the increment in tension in the ring will be $\frac{Qq}{n\pi^2\varepsilon_0 r^2}$. Find n.

PEO282

For More Join: @IITJEE_Advanced

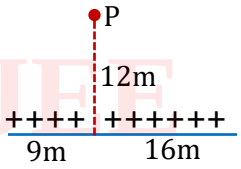
Electric Field

7. Two charges are separated as shown in the diagram *A* below. Each charge has a magnitude of $50nC$ in diagram *A*. If the charge on the right is moved further to the right as depicted in diagram *B*, additional charge that must be removed from the left in order to maintain the initial electric field magnitude at the center (in nC) is given by $\frac{\alpha}{9}nC$. Fill the value of α in OMR sheet.



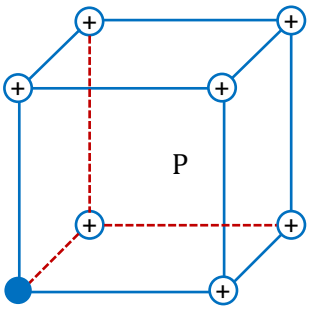
PEO102

8. Find the magnitude of electric field (in N/C) due to a line charge of $\lambda = \left(2\sqrt{2}\right) nC/m$ at a point *P* as shown.



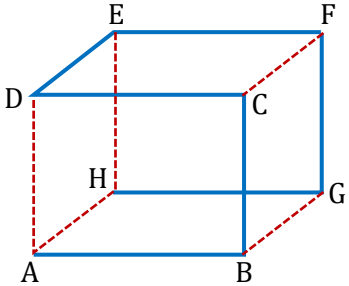
PEO124

9. Consider a regular cube with positive point charge $+Q$ in all corners except for one which has a negative point charge $-Q$. Let the distance from any corner to the center of the cube be *r*. The magnitude of electric field at point *P*, the center of the cube is $\frac{nkQ}{r^2}$. Find *n*?



PEO283

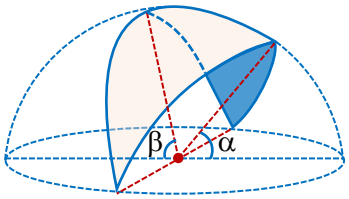
10. 7 rods, *AB*, *BC*, *CD*, *DE*, *EF*, *FG* and *GH* are uniformly charged. Electric field due to each one is *E* at the centre of cube. Net electric field due to the entire combination is $\sqrt{n}E$. Find *n*.



PEO284

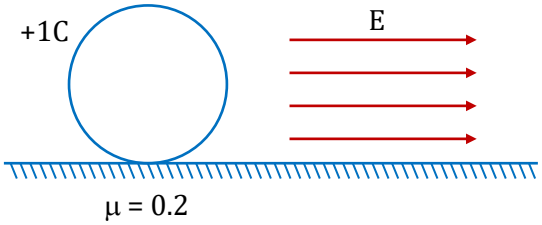
For More Join: @IITJEE_Advanced

11. The electric field intensity at the centre of a uniformly charged hemispherical shell is E_0. Now two portions of the hemisphere are cut from either side and remaining portion is shown in figure. If $\alpha = \beta = \pi/3$, then electric field intensity at centre due to remaining portion is $\frac{E_0}{n}$. Find n.



PEO285

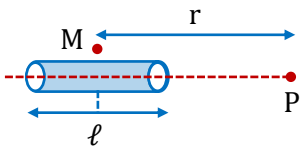
12. A non-conducting disc of mass = $2kg$, total charge = $+1C$ uniformly distributed, is placed on a rough horizontal non-conducting surface with its cross-section in vertical plane as shown. A uniform horizontal electric field E is now switched on. Find maximum magnitude of electric field E in (N/C) so that the disc rolls purely. [$g = 10\ ms^{-2}$]



PEO286

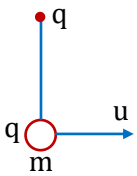
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13. A insulating cylindrical rod of diameter d and length ℓ ($\ell >> d$) has a uniform surface charge density such that the electric field just outside the curved surface of the cylinder at point M is E_0. The electric field due to charge distribution at point P ($r >> \ell$) is $E_0 \frac{\ell d}{nr^2}$. Find n.



PEO287

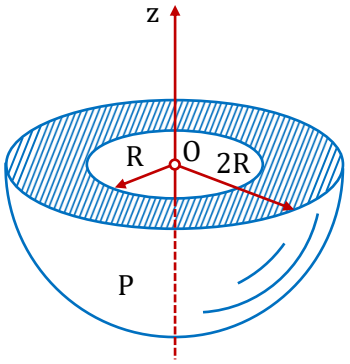
14. The bob of a pendulum has mass $m = 1\ kg$ and charge $q = +40\ \mu C$. Length of pendulum is $\ell = 0.9\ m$. The point of suspension also has the same charge $+40\ \mu C$. What the minimum speed u (in m/s) should be imparted to the bob so that it can complete vertical circle ?



PEO288

For More Join: @IITJEE_Advanced

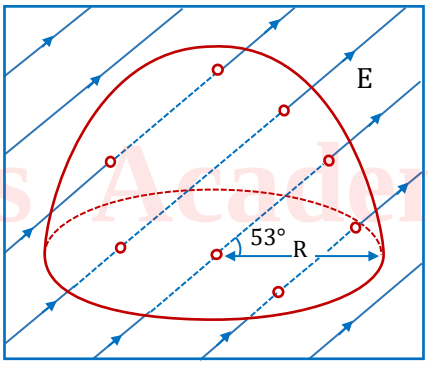
15. A hemispherical bowl of inner radii R and outer radii $2R$ is charged with density $\rho=\rho_o\frac{r}{R}$. The electric field at point O *is* $\frac{n\rho_o R}{8\varepsilon_o}$. Find n.



PEO289

Properties of Electric Line of Force and Gauss Law

16. A uniform electric field $E = 500\ N/C$ passes through a hemispherical surface of radius $R = 1.2\ m$ as shown in figure. The net electric flux (in S.I. units) through the hemispherical surface only is $N\pi$. Then find the value of N.



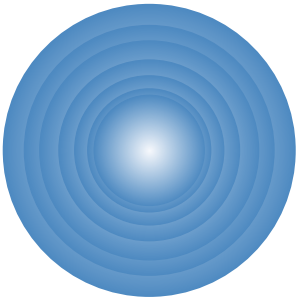
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PEO139

17. A spherical cavity of radius $2m$ is made whose centre is at a distance of $2m$ from centre of a uniformly charged solid sphere of density $\rho = 8.85 \times 10^{-12}\ C/m^3$ and radius $5m$. If electric field inside cavity is given by $x/y\ N/C$, fill value of $(y\text{-}x)$. (y and x are minimum possible integers).

PEO176

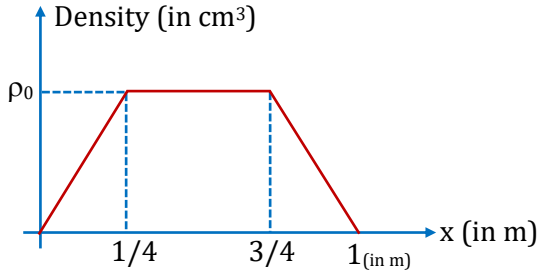
18. In space of a spherical charge distribution with radial charge density $\rho(r)=\rho_0\frac{e^{-r/2a}}{r^2}$. Total charge in the distribution is Q. The value of $\frac{Q}{\rho_0\pi a}$ is :



PEO290

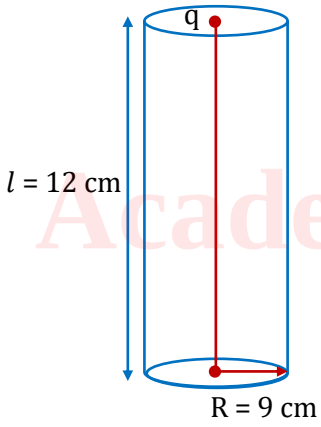
For More Join: @IITJEE_Advanced

19. The volume charge density as a function of distance X from one face inside a unit cube is varying as shown in the figure. The total flux (in S.I. units) through the cube if ($\rho_0 = 8.85 \times 10^{-12}\ C/m^3$) is $\frac{n}{4}$, Find n?



PEO291

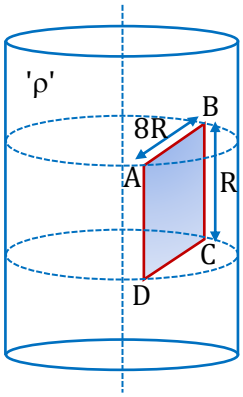
20. A point charge is placed at the centre of top circular end of a cylinder of length $l = 12cm$ and radius $R = 9\ cm$ as shown. Electric flux through the curved surface of the cylinder is $\frac{nq}{5\varepsilon_0}$: Find n ?



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PEO292

21. A non-conducting solid cylinder of infinite length having charge density ρ and radius of cylinder is $5R$. If the flux passing through the surface $ABCD$ as shown in figure is $\frac{2\theta\rho R^3}{\varepsilon_0}$. Find the value of θ.



PEO293

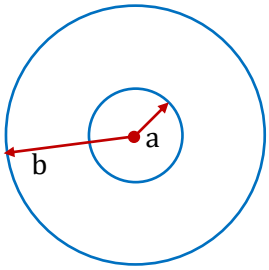
For More Join: @IITJEE_Advanced

Electric Potential Energy and Electric Potential

22. A particle of mass m carrying charge 'q' is projected with velocity (v) from point P towards an infinite line charge from a distance 'a'. Its speed reduces to zero momentarily at point Q which is at a distance $a/2$ from the line charge. If another particle with mass m and charge $-q$ is projected with the same velocity v from point P towards the line charge. Its speed is found to be $\frac{Nv}{\sqrt{2}}$ at point 'Q'. Find the value of N.

PEO218

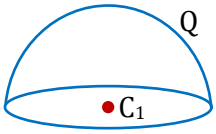
23. Outer cylinder of the coaxial non-conductor of radius 'b' is given a positive potential V relative to the inner cylinder of radius 'a' as shown in the figure (charge distribution is uniform). A charge q (mass = m) is set free with negligible velocity at the surface of the inner cylinder. Find the velocity (in m/s), when it hits the outer cylinder. [Consider $V = 10$, $q = -20$, $m = 1$ all in S.I. Units]



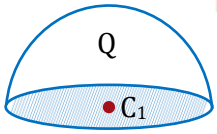
PEO219

24. Charges Q each are distributed uniformly on a nonconducting hemispherical shell (on the surface) and in the volume of a nonconducting solid hemisphere. A point charge is brought at the centre C_1 of shell and at the centre C_2 of solid hemisphere from infinity. If work done in these process are W_1 and W_2 the W_1/W_2 is equal $n/3$. Find n?

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Hollow (without base)

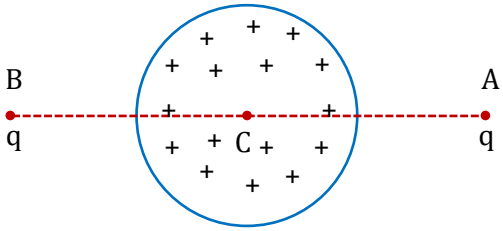


Solid hemisphere

PEO294

25. A uniformly charged non-conducting sphere having charge Q and radius R is kept in gravity free space. Two identical charge particles are released from rest from 'A' and 'B' simultaneously. The speed of one of the charge particle when separation between them is very large, is $\sqrt{\frac{Kq}{nmR}\left(\frac{q}{2}+2Q\right)}$ (mass of one of the particle is m), Find n.

[$C \rightarrow$ centre of sphere and A, B and C in same line. $AC = BC = 2R$, $K = \frac{1}{4\pi\varepsilon_0}$]



PEO295

For More Join: @IITJEE_Advanced

26. Consider the potential at the corner of a uniformly charged cube of dimension L to be directly proportional to ρL^2, where ρ is volume charge density. The value of the ratio of the potential at the centre to the potential at the corner of the cube is n:1. Find n?

PEO296

27. A charge q is uniformly distributed on the surface of a disc of radius R. Then a circular hole of radius $R/2$ is punched taking one of the radii as the diameter of the hole. The potential due to remaining portion of the disc at the center of the disc is nV_0. $\left[V_0 = \frac{q}{2\pi\varepsilon_0 R}\left(1-\frac{1}{\pi}\right)\right]$. Find n.

PEO297

Dipole

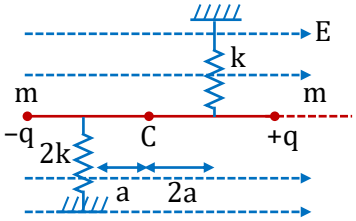
28. A plastic rod of length 1.0 m carries uniform positive charge +4.0 μC on half of its length and uniform negative charge –4.0 μC on the remaining half of its length. Find magnitude of it's net dipole moment in μC–m.



PEO232

29. Two particles of mass 'm' each and having charge $+q$ and $-q$ respectively joined by a light rod of length $6a$ as shown in the figure. The whole system is kept on a horizontal frictionless surface and two spring are connected to the rod. Rod is hinged at centre and assume no energy loss takes place in form of electromagnetic wave during the motion of particles. A uniform electric field exists in the region as shown in the figure in entire region. The rod is given a small angular displacement from equilibrium, then the frequency of small oscillation is v then fill value of $4v$.

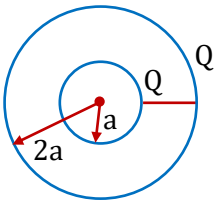
$\left[\text{Given } m = \frac{k}{3\pi^2}kg \text{ and } Eq = 8ka\right]$



PEO298

Conductors

30. Figure shows a solid metal sphere of radius a surrounded by a concentric thin metal shell of radius $2a$. Initially both are having charges Q each. When the two are connected by a conducting wire as shown in figure, then amount of heat produced in this process will be $\frac{KQ^2}{\lambda a}$. Find the value of λ.



PEO267

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. An electric dipole has a fixed dipole moment $\vec{p}$, which makes angle θ with respect to *x*-axis. When subjected to an electric field $\vec{E}_1 = E\hat{i}$, it experiences a torque $\vec{T}_1 = \tau\hat{k}$. When subjected to another electric field $\vec{E}_2 = \sqrt{3}E_1\hat{j}$ it experiences torque $\vec{T}_2 = -\vec{T}_1$. The angle θ is : **[JEE (Main) 2017]**

(1) 60° (2) 90° (3) 30° (4) 45°

PEO024

2. Three concentric metal shells *A*, *B* and *C* of respective radii *a*, *b* and *c* ($a < b < c$) have surface charge densities +σ, −σ and +σ respectively. The potential of shell *B* is **[JEE (Main) 2018]**

(1) $\frac{\sigma}{\varepsilon_0}\left[\frac{a^2-b^2}{b}+c\right]$ (2) $\frac{\sigma}{\varepsilon_0}\left[\frac{b^2-c^2}{b}+a\right]$ (3) $\frac{\sigma}{\varepsilon_0}\left[\frac{b^2-c^2}{c}+a\right]$ (4) $\frac{\sigma}{\varepsilon_0}\left[\frac{a^2-b^2}{a}+c\right]$

PEO025

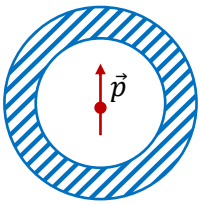
3. Charge is distributed within a sphere of radius *R* with a volume charge density $\rho(r)=\frac{A}{r^2}e^{-2r/a}$, where *A* and *a* are constants. If *Q* is the total charge of this charge distribution, the radius *R* is : **[JEE (Main) 2019]**

(1) $\frac{a}{2}\log\left(1-\frac{Q}{2\pi aA}\right)$ (2) $a\log\left(1-\frac{Q}{2\pi aA}\right)$ (3) $a\log\left(\frac{1}{1-\frac{Q}{2\pi aA}}\right)$ (4) $\frac{a}{2}\log\left(\frac{1}{1-\frac{Q}{2\pi aA}}\right)$

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PEO026

4. Shown in the figure is a shell made of a conductor. It has inner radius *a* and outer radius *b*, and carries charge *Q*. At its centre is a dipole $\vec{p}$ as shown. In this case **[JEE (Main) 2019]**



(1) Electric field outside the shell is the same as that of a point charge at the centre of the shell.

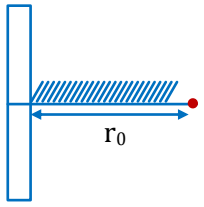
(2) Surface charge density on the inner surface of the shell is zero everywhere

(3) Surface charge density on the inner surface is uniform and equal to $\frac{(Q/2)}{4\pi a^2}$

(4) Surface charge density on the outer surface depends on $|\vec{p}|$

PEO030

5. A positive point charge is released from rest at a distance r_0 from a positive line charge with uniform density. The speed (*v*) of the point charge, as a function of instantaneous distance *r* from line charge, is proportional to: **[JEE (Main) 2019]**



(1) $v \propto e^{+r/r_0}$ (2) $v \propto \ell n\left(\frac{r}{r_0}\right)$

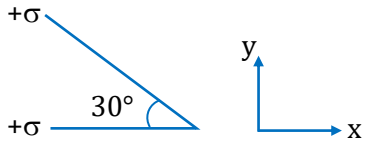
(3) $v \propto \left(\frac{r}{r_0}\right)$ (4) $v \propto \sqrt{\ell n\left(\frac{r}{r_0}\right)}$

PEO300

For More Join: @IITJEE_Advanced

6. Two infinite planes each with uniform surface charge density + σ are kept in such a way that the angle between them is 30°. The electric field in the region shown between them is given by:

[JEE (Main) 2020]



(1) $\frac{\sigma}{\varepsilon_0}\left[\left(1+\frac{\sqrt{3}}{2}\right)\hat{y}+\frac{\hat{x}}{2}\right]$

(2) $\frac{\sigma}{2\varepsilon_0}\left[\left(1-\frac{\sqrt{3}}{2}\right)\hat{y}-\frac{\hat{x}}{2}\right]$

(3) $\frac{\sigma}{2\varepsilon_0}\left[\left(1+\sqrt{3}\right)\hat{y}+\frac{\hat{x}}{2}\right]$

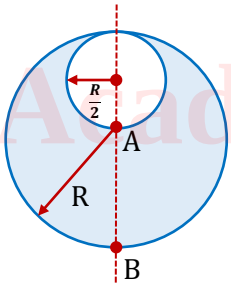
(4) $\frac{\sigma}{2\varepsilon_0}\left[\left(1+\sqrt{3}\right)\hat{y}-\frac{\hat{x}}{2}\right]$

PEO031

7. Consider a sphere of radius R which carries a uniform charge density ρ. If a sphere of radius $\frac{R}{2}$ is carved out of it, as shown, the ratio $\frac{|\vec{E}_A|}{|\vec{E}_B|}$ of magnitude of electric field $\vec{E}_A$ and $\vec{E}_B$, respectively, at points A and B due to the remaining portion is:

[JEE (Main) 2020]

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(1) $\frac{18}{54}$

(2) $\frac{21}{34}$

(3) $\frac{17}{54}$

(4) $\frac{18}{34}$

PEO032

8. A solid sphere of radius R carries a charge $(Q + q)$ distributed uniformly over its volume. A very small point like piece of it of mass m gets detached from the bottom of the sphere and falls down vertically under gravity. This piece carries charge q. If it acquires a speed v when it has fallen through a vertical height y (see figure), then :

(Assume the remaining portion to be spherical).

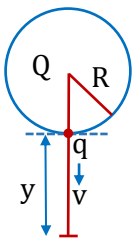
[JEE (Main) 2020]

(1) $v^2 = 2y\left[\frac{qQ}{4\pi\in_0 R(R+y)m}+g\right]$

(2) $v^2 = y\left[\frac{qQ}{4\pi\in_0 R^2 ym}+g\right]$

(3) $v^2 = 2y\left[\frac{qQR}{4\pi\in_0 (R+y)^3 m}+g\right]$

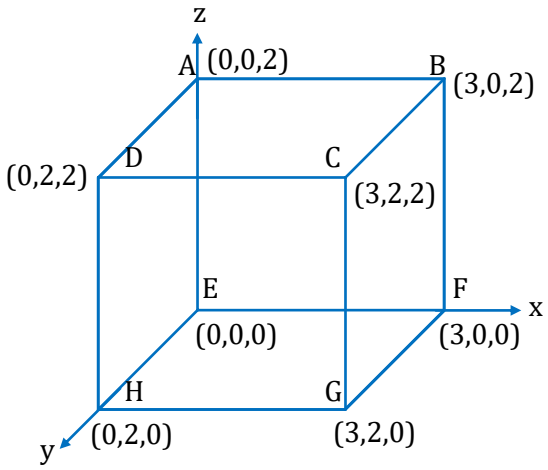
(4) $v^2 = y\left[\frac{qQ}{4\pi\in_0 R(R+y)m}+g\right]$



PEO034

For More Join: @IITJEE_Advanced

9. An electric field $\vec{E}=4x\hat{i}-(y^2+1)\hat{j}\,N/C$ passes through the box shown in figure. The flux of the electric field through surfaces ABCD and BCGF are marked as ϕ_I and ϕ_{II} respectively. The difference between $(\phi_I - \phi_{II})$ is (in Nm^2/C) _____. **[JEE (Main) 2020]**



PEO301

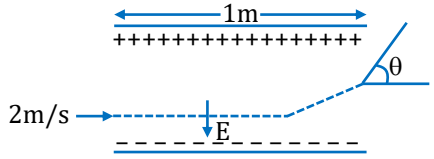
10. Two identical charged particles each having a mass 10 g and charge 2.0×10^{-7} C area placed on a horizontal table with a separation of L between then such that they stay in limited equilibrium. If the coefficient of friction between each particle and the table is 0.25, find the value of L.
[Use g = 10 ms^{-2}] **[JEE (Main) 2022]**

(1) 12 cm (2) 10 cm (3) 8 cm (4) 5 cm

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PEO037

11. A uniform electric field $E = (8m/e)\ V/m$ is created between two parallel plates of length $1m$ as shown in figure, (where m = mass of electron and e = charge of electron). An electron enters the field symmetrically between the plates with a speed of $2m/s$. The angle of the deviation (θ) of the path of the electron as it comes out of the field will be _____. **[JEE (Main) 2022]**



(1) $\tan^{-1}(4)$ (2) $\tan^{-1}(2)$ (3) $\tan^{-1}\left(\frac{1}{3}\right)$ (4) $\tan^{-1}(3)$

PEO302

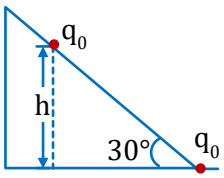
12. A dipole comprises of two charged particles of identical magnitude q and opposite in nature. The mass 'm' of the positive charged particle is half of the mass of the negative charged particle. The two charges are separated by a distance 'l'. If the dipole is placed in a uniform electric field '$\vec{E}$'; such a way that dipole axis makes a very small angle with the electric field, '$\vec{E}$'. The angular frequency of the oscillations of the dipole when released is given by: **[JEE (Main) 2023]**

(1) $\sqrt{\frac{8qE}{3ml}}$ (2) $\sqrt{\frac{4qE}{ml}}$ (3) $\sqrt{\frac{4qE}{3ml}}$ (4) $\sqrt{\frac{8qE}{ml}}$

PEO268

For More Join: @IITJEE_Advanced

13. As shown in the figure. a configuration of two equal point charges ($q_0 = \pm 2\mu C$) is placed on an inclined plane. Mass of each point charge is 20 g. Assume that there is no friction between charge and plane. For the system of two point charges to be in equilibrium (at rest) the height $h = x \times 10^{-3}\ m$. The value of x is _____.



(Take: $\frac{1}{4\pi\varepsilon_0} = 9\times10^9\ N\,m^2C^{-2}, g = 10\,ms^{-1}$) **[JEE (Main) 2023]**

PEO303

14. An electron is moving under the influence of the electric field of a uniformly charged infinite plane sheet S having surface charge density $+\sigma$. The electron at $t = 0$ is at a distance of 1 m from S and has a speed of 1 m/s. The maximum value of σ if the electron strikes S at $t = 1\ s$ is $\alpha\left[\frac{m\in_0}{e}\right]\frac{C}{m^2}$ the value of α is **[JEE (Main) 2024]**

PEO304

15. Two charges of $5Q$ and $-2Q$ are situated at the points $(3a, 0)$ and $(-5a, 0)$ respectively. The electric flux through a sphere of radius '$4a$' having center at origin is :

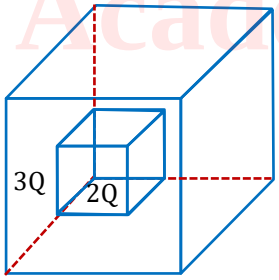
[JEE (Main) 2024]

(1) $\frac{2Q}{\varepsilon_0}$ (2) $\frac{5Q}{\varepsilon_0}$ (3) $\frac{7Q}{\varepsilon_0}$ (4) $\frac{3Q}{\varepsilon_0}$

PEO305

16. C_1 and C_2 are two hollow concentric cubes enclosing charges $2Q$ and $3Q$ respectively as shown in figure. The ratio of electric flux passing through C_1 and C_2 is : **[JEE (Main) 2024]**

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(1) 2 : 5 (2) 5 : 2 (3) 2 : 3 (4) 3 : 2

PEO306

17. Given below are two statements : one is labelled a Assertion (A) and the other is labelled as Reason(R)

Assertion (A) : Work done by electric field on moving a positive charge on an equipotential surface is always zero.

Reason (R) : Electric lines of forces are always perpendicular to equipotential surfaces.

In the light of the above statements, choose the most appropriate answer from the options given below :

[JEE (Main) 2024]

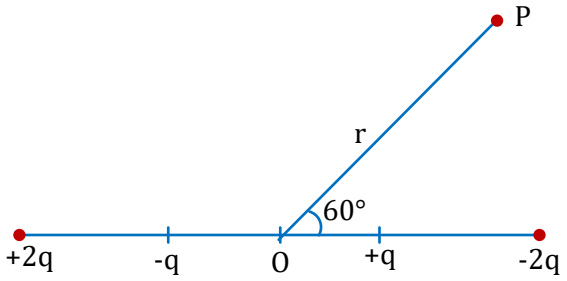
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(2) (A) is correct but (R) is not correct
(3) (A) is not correct but (R) is correct
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

PEO307

For More Join: @IITJEE_Advanced

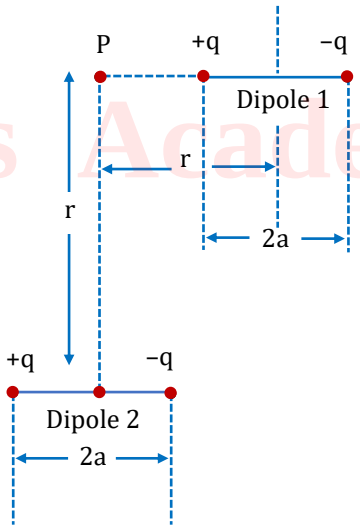
18. The distance between charges $+q$ and $-q$ is $2l$ and between $+2\,q$ and $-2\,q$ is $4l$. The electrostatic potential at point P at a distance r from centre O is $-\alpha\left[\frac{ql}{r^2}\right]\times10^9 V$, where the value of α is _____.

(Use $\frac{1}{4\pi\varepsilon_0}=9\times10^9\ Nm^2C^{-2}$) **[JEE (Main) 2024]**



PEO308

19. A point particle of charge Q is located at P along the axis of an electric dipole 1 at a distance r as shown in the figure. The point P is also on the equatorial plane of a second electric dipole 2 at a distance r. The dipoles are made of opposite charge q separated by a distance $2a$. For the charge particle at P not to experience any net force, which of the following correctly describes the situation? **[JEE (Main) 2025]**



(1) $\frac{a}{r}\sim20$ (2) $\frac{a}{r}\sim10$ (3) $\frac{a}{r}\sim0.5$ (4) $\frac{a}{r}\sim3$

PEO393

20. Three infinitely long wires with linear charge density λ are placed along the x-axis, y-axis and z-axis respectively. Which of the following denotes an equipotential surface ?

[JEE (Main) 2025]

(1) $xy + yz + zx$ = constant (2) $(x + y)(y + z)(z + x)$ = constant

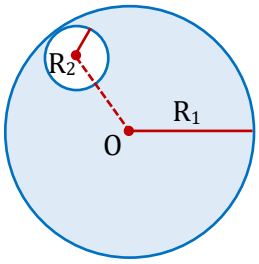
(3) $(x^2 + y^2)(y^2 + z^2)(z^2 + x^2)$ = constant (4) xyz = constant

PEO394

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. Consider a uniform spherical distribution of radius R_1 centred at the origin O. In this distribution, a spherical cavity of radius R_2, centred at P with distance $OP = a = R_1 - R_2$ (see figure) is made. If the electric field inside the cavity at position $\vec{r}$ is $\vec{E}(\vec{r})$, then the correct statement(s) is(are) :



[JEE (Advanced) 2015]

(A) $\vec{E}$ is uniform, its magnitude is independent of R_2 but its direction depends on $\vec{r}$

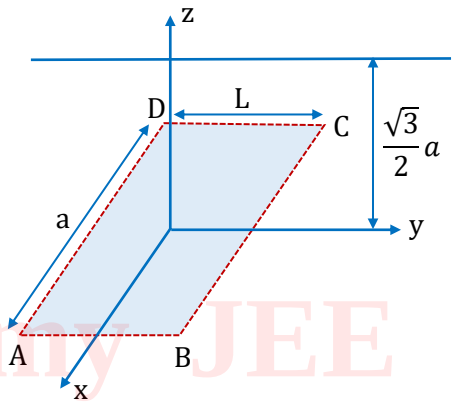
(B) $\vec{E}$ is uniform, its magnitude depends on R_2 and its direction depends on $\vec{r}$

(C) $\vec{E}$ is uniform, its magnitude is independent of a but its direction depends on $\vec{a}$

(D) $\vec{E}$ is uniform and both its magnitude and direction depend on $\vec{a}$

PEO054

2. An infinitely long uniform line charge distribution of charge per unit length λ lies parallel to the y-axis in the y-z plane at $z = \frac{\sqrt{3}}{2}a$ (see figure). If the magnitude of the flux of the electric field through the rectangular surface $ABCD$ lying in the x-y plane with its centre at the origin is $\frac{\lambda L}{n\varepsilon_0}$ (ε_0 = permittivity of free space) then the value of n is.



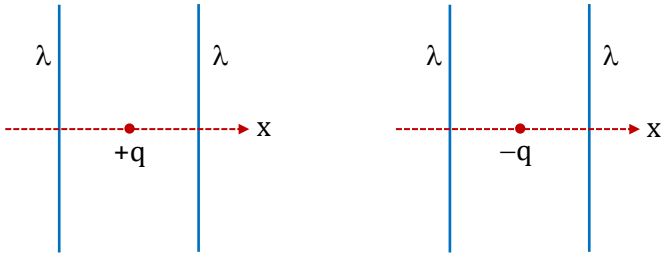
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[JEE (Advanced) 2015]

PEO055

3. The figures below depict two situations in which two infinitely long static line charges of constant positive line charge density λ are kept parallel to each other. In their resulting electric field, point charges q and $-q$ are kept in equilibrium between them. The point charges are confined to move in the x direction only. If they are given a small displacement about their equilibrium positions, then the correct statement(s) is (are) :

[JEE (Advanced) 2015]



(A) Both charges execute simple harmonic motion.

(B) Both charges will continue moving in the direction of their displacement.

(C) Charge $+q$ executes simple harmonic motion while charge $-q$ continues moving in the direction of its displacement.

(D) Charge $-q$ executes simple harmonic motion while charge $+q$ continues moving in the direction of its displacement.

PEO056

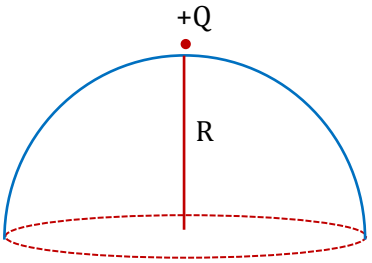
For More Join: @IITJEE_Advanced

4. A length-scale (ℓ) depends on the permittivity (ε) of a dielectric material, Boltzmann constant k_B, the absolute temperature T, the number per unit volume (n) of certain charged particles, and the charge (q) carried by each of the particles. Which of the following expressions(s) for ℓ is(are) dimensionally correct? **[JEE (Advanced) 2016]**

(A) $\ell = \sqrt{\left(\frac{nq^2}{\varepsilon k_B T}\right)}$ (B) $\ell = \sqrt{\left(\frac{\varepsilon k_B T}{nq^2}\right)}$ (C) $\ell = \sqrt{\left(\frac{q^2}{\varepsilon n^{2/3} k_B T}\right)}$ (D) $\ell = \sqrt{\left(\frac{q^2}{\varepsilon n^{1/3} k_B T}\right)}$

PEO057

5. A point charge $+Q$ is placed just outside an imaginary hemispherical surface of radius R as shown in the figure. Which of the following statements is/are correct ? **[JEE (Advanced) 2017]**



(A) The circumference of the flat surface is an equipotential

(B) The electric flux passing through the curved surface of the hemisphere is $-\frac{Q}{2\varepsilon_0}\left(1-\frac{1}{\sqrt{2}}\right)$

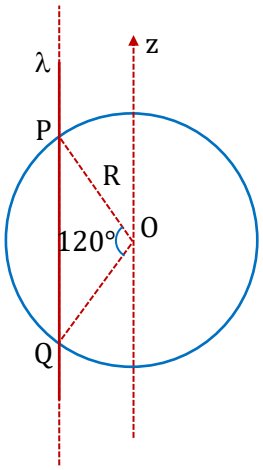
(C) Total flux through the curved and the flat surfaces is $\frac{Q}{\varepsilon_0}$

(D) The component of the electric field normal to the flat surface is constant over the surface.

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PEO058

6. An infinitely long thin non-conducting wire is parallel to the z-axis and carries a uniform line charge density λ. It pierces a thin non-conducting spherical shell of radius R in such a way that the arc PQ subtends an angle 120° at the centre O of the spherical shell, as shown in the figure. The permittivity of free space is ε_0. Which of the following statements is (are) true ? **[JEE (Advanced) 2018]**



(A) The electric flux through the shell is $\sqrt{3}R\lambda/\varepsilon_0$.

(B) The z-component of the electric field is zero at all the points on the surface of the shell.

(C) The electric flux through the shell is $\sqrt{2}R\lambda/\varepsilon_0$.

(D) The electric field is normal to the surface of the shell at all points.

PEO059

7. A particle, of mass 10^{-3} kg and charge 1.0 C, is initially at rest. At time $t = 0$, the particle comes under the influence of an electric field $\vec{E}(t) = E_0 \sin \omega t \hat{\imath}$ where $E_0 = 1.0$ $N\,C^{-1}$ and $\omega = 10^3$ rad s^{-1}. Consider the effect of only the electrical force on the particle. Then the maximum speed, in ms^{-1}, attained by the particle at subsequent times is **[JEE (Advanced) 2018]**

PEO060

For More Join: @IITJEE_Advanced

8. The electric field E is measured at a point $P(0, 0, d)$ generated due to various charge distributions and the dependence of E on d is found to be different for different charge distributions. List-I contains different relations between E and d. List-II describes different electric charge distributions, along with their locations. Match the functions in List-I with the related charge distributions in List-II. **[JEE (Advanced) 2018]**

	List–I		**List–II**
P.	E is independent of d	1.	A point charge Q at the origin
Q.	$E \propto \frac{1}{d}$	2.	A small dipole with point charges Q at $(0,0,\ell)$ and $-Q$ at $(0,0,-\ell)$. Take $2\ell << d$
R.	$E \propto \frac{1}{d^2}$	3.	An infinite line charge coincident with the x-axis, with uniform linear charge density λ.
S.	$E \propto \frac{1}{d^3}$	4.	Two infinite wires carrying uniform linear Charge density parallel to the x-axis. The one along $(y = 0, z = \ell)$ has a charge density $+\lambda$ and the one along $(y = 0, z = -\ell)$ has a charge density $-\lambda$. Take $2\ell << d$
		5.	Infinite plane charge coincident with the xy-plane with uniform surface charge density

(A) $P \to 5$; $Q \to 3, 4$; $R \to 1$; $S \to 2$
(B) $P \to 5$; $Q \to 3,$; $R \to 1,4$; $S \to 2$
(C) $P \to 5$; $Q \to 3,$; $R \to 1,2$; $S \to 4$
(D) $P \to 4$; $Q \to 2, 3$; $R \to 1$; $S \to 5$

PEO061

9. A thin spherical insulating shell of radius R carries a uniformly distributed charge such that the potential at its surface is V_0. A hole with a small area $\alpha 4\pi R^2$ ($\alpha << 1$) is made on the shell without affecting the rest of the shell. Which one of the following statements is correct ?

[JEE (Advanced) 2019]

(A) The ratio of the potential at the centre of the shell to that of the point at $\frac{1}{2}R$ from centre towards the hole will be $\frac{1-\alpha}{1-2\alpha}$

(B) The magnitude of electric field at the center of the shell is reduced by $\frac{\alpha V_0}{2R}$

(C) The magnitude of electric field at a point, located on a line passing through the hole and shell's center on a distance $2R$ from the center of the spherical shell will be reduced by $\frac{\alpha V_0}{2R}$

(D) The potential at the center of the shell is reduced by $2\alpha V_0$

PEO062

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For More Join: @IITJEE_Advanced

10. A charged shell of radius R carries a total charge Q. Given Φ as the flux of electric field through a closed cylindrical surface of height h, radius r and with its center same as that of the shell. Here, center of the cylinder is a point on the axis of the cylinder which is equidistant from its top and bottom surfaces. Which of the following option(s) is/are correct ?
[$\in_0$ is the permittivity of free space] **[JEE (Advanced) 2019]**

(A) If $h > 2R$ and $r > R$ then $\Phi = \frac{Q}{\in_0}$

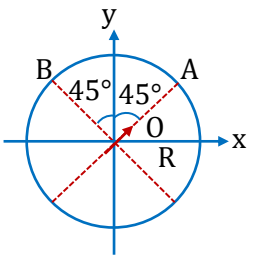
(B) If $h < \frac{8R}{5}$ and $r = \frac{3R}{5}$ then $\Phi = 0$

(C) If $h > 2R$ and $r = \frac{4R}{5}$ then $\Phi = \frac{Q}{5\in_0}$

(D) If $h > 2R$ and $r = \frac{3R}{5}$ then $\Phi = \frac{Q}{5\in_0}$

PEO063

11. An electric dipole with dipole moment $\frac{p_0}{\sqrt{2}}(\hat{i}+\hat{j})$ is held fixed at the origin O in the presence of an uniform electric field of magnitude E_0. If the potential is constant on a circle of radius R centered at the origin as shown in figure, then the correct statement(s) is/are:
(ε_0 is permittivity of free space, $R >>$ dipole size) **[JEE (Advanced) 2019]**



(A) $R = \left(\frac{p_0}{4\pi\varepsilon_0 E_0}\right)^{1/3}$

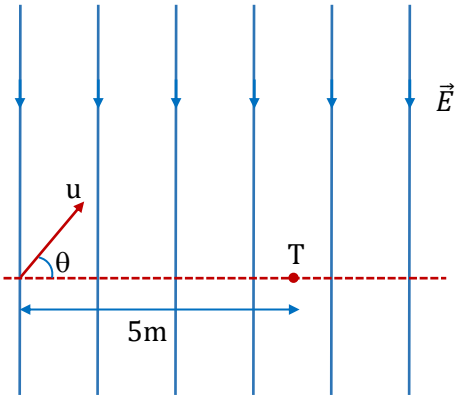
(B) The magnitude of total electric field on any two points of the circle will be same

(C) Total electric field at point A is $\vec{E}_A = \sqrt{2}E_0(\hat{i}+\hat{j})$

(D) Total electric field at point B is $\vec{E}_B = 0$

PEO064

12. A uniform electric field, $\vec{E} = -400\sqrt{3}\hat{y}\ NC^{-1}$ is applied in a region. A charged particle of mass m carrying positive charge q is projected in this region with an initial speed of $2\sqrt{10} \times 10^6\ ms^{-1}$. This particle is aimed to hit a target T, which is $5m$ away from its entry point into the field as shown schematically in the figure. Take $\frac{q}{m} = 10^{10}\ Ckg^{-1}$. Then: **[JEE (Advanced) 2020]**



(A) the particle will hit T if projected at an angle 45° from the horizontal

(B) the particle will hit T if projected either at an angle 30° or 60° from the horizontal

(C) time taken by the particle to hit T could be $\sqrt{\frac{5}{6}}$ μs as well as $\sqrt{\frac{5}{2}}$ μs

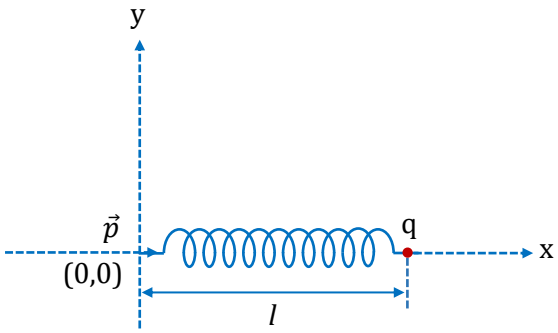
(D) time taken by the particle to hit T is $\sqrt{\frac{5}{3}}$ μs

PEO065

For More Join: @IITJEE_Advanced

13. One end of a spring of negligible unstretched length and spring constant k is fixed at the origin (0, 0). A point particle of mass m carrying a positive charge q is attached at its other end. The entire system is kept on a smooth horizontal surface. When a point dipole $\vec{p}$ pointing towards the charge q is fixed at the origin, the spring gets stretched to a length ℓ and attains a new equilibrium position (see figure below). If the point mass is now displaced slightly by $\Delta\ell << \ell$ from its equilibrium position and released, it is found to oscillate at frequency $\frac{1}{\delta}\sqrt{\frac{k}{m}}$. The value of δ is.

[JEE (Advanced) 2020]



PEO066

14. A circular disc of radius R carries surface charge density $\sigma(r) = \sigma_0\left(1-\frac{r}{R}\right)$, where σ_0 is a constant and r is the distance from the center of the disc. Electric flux through a large spherical surface that encloses the charged disc completely is ϕ_0. Electric flux through another spherical surface of radius $\frac{R}{4}$ and concentric with the disc is ϕ. Then the ratio $\frac{\phi_0}{\phi}$ is_____. **[JEE (Advanced) 2020]**

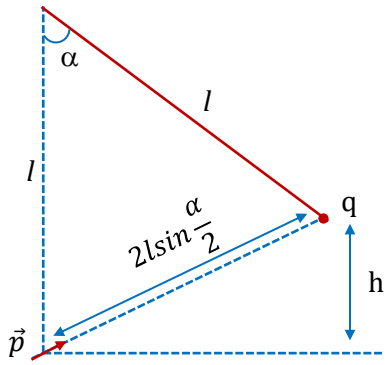
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PEO067

15. A point charge q of mass m is suspended vertically by a string of length ℓ. A point dipole of dipole moment $\vec{p}$ is now brought towards q from infinity so that the charge moves away. The final equilibrium position of the system including the direction of the dipole, the angles and distances is shown in the figure below. If the work done in bringing the dipole to this position is $N \times (mgh)$, where g is the acceleration due to gravity, then the value of N is _____.

(Note that for three coplanar forces keeping a point mass in equilibrium, $\frac{F}{\sin\theta}$ is the same for all forces, where F is any one of the forces and θ is the angle between the other two forces)

[JEE (Advanced) 2020]



PEO068

For More Join: @IITJEE_Advanced

16. Two identical non-conducting solid spheres of same mass and charge are suspended in air from a common point by two non-conducting, massless strings of same length. At equilibrium, the angle between the strings is α. The spheres are now immersed in a dielectric liquid of density 800 $kg\ m^{-3}$ and dielectric constant 21. If the angle between the strings remains the same after the immersion, then **[JEE (Advanced) 2020]**

(A) electric force between the spheres remains unchanged

(B) electric force between the spheres reduces

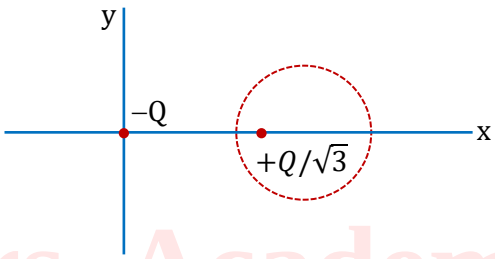
(C) mass density of the spheres is 840 $kg\ m^{-3}$

(D) the tension in the strings holding the spheres remains unchanged

PEO069

PARAGRAPH FOR QUESTIONS NO. 17 AND 18

Two points charges $-Q$ and $+Q\sqrt{3}$ are placed in the xy-plane at the origin (0, 0) and a point (2,0), respectively, as shown in the figure. This results in an equipotential circle of radius R and potential $V = 0$ in the xy-plane with its center at $(b, 0)$. All lengths are measured in meters. **[JEE (Advanced) 2021]**



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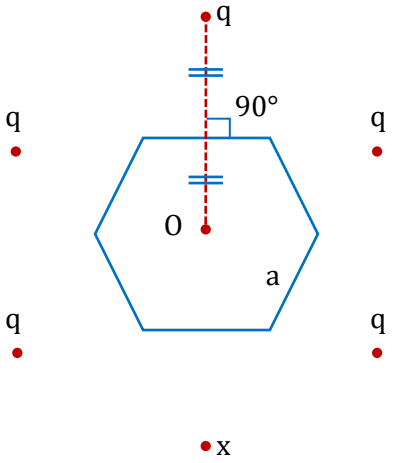
17. The value of R is ___ meter.

PEO070

18. The value of b is ___ meter.

PEO071

19. Six charges are placed around a regular hexagon of side length a as shown in the figure. Five of them have charge q, and the remaining one has charge x. The perpendicular from each charge to the nearest hexagon side passes through the center O of the hexagon and is bisected by the side. **[JEE (Advanced) 2022]**



Which of the following statement(s) is(are) correct in SI units?

(A) When $x = q$, the magnitude of the electric field at O is zero.

(B) When $x = -q$, the magnitude of the electric field at O is $\frac{q}{6\pi\epsilon_0 a^2}$.

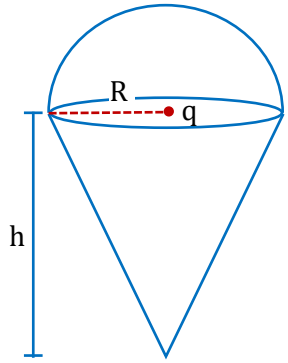
(C) When $x = 2q$, the potential at O is $\frac{7q}{4\sqrt{3}\pi\epsilon_0 a}$.

(D) When $x = -3q$, the potential at O is $\frac{3q}{4\sqrt{3}\pi\epsilon_0 a}$.

PEO072

For More Join: @IITJEE_Advanced

20. A charge q is surrounded by a closed surface consisting of an inverted cone of height h and base radius R, and a hemisphere of radius R as shown in the figure. The electric flux through the conical surface is $\frac{nq}{6\epsilon_0}$ (in SI units). The value of n is _____. **[JEE (Advanced) 2022]**



PEO073

21. In the figure, the inner (shaded) region A represents a sphere of radius $r_A = 1$, within which the electrostatic charge density varies with the radial distance r from the center as $\rho_A = kr$, where k is positive. In the spherical shell B of outer radius r_B, the electrostatic charge density varies as $\rho_B = \frac{2k}{r}$. Assume that dimensions are taken care of. All physical quantities are in their SI units.

[JEE (Advanced) 2022]

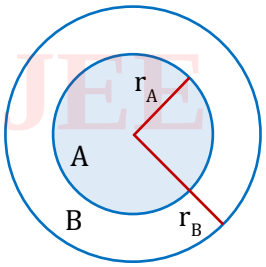
Which of the following statement(s) is(are) correct?

(A) If $r_B = \sqrt{\frac{3}{2}}$, then the electric field is zero everywhere outside B.

(B) If $r_B = \frac{3}{2}$, then the electric potential just outside B is $\frac{k}{\epsilon_0}$.

(C) If $r_B = 2$, then the total charge of the configuration is $15\pi k$.

(D) If $r_B = \frac{5}{2}$, then the magnitude of the electric field just outside B is $\frac{13\pi k}{\epsilon_0}$.



PEO074

22. A disk of radius R with uniform positive charge density σ is placed on the xy plane with its center at the origin. The Coulomb potential along the z-axis is $V(z) = \frac{\sigma}{2\epsilon_0}\left(\sqrt{R^2 + z^2} - z\right)$. A particle of positive charge q is placed initially at rest at a point on the z axis with $z = z_0$ and $z_0 > 0$. In addition to the Coulomb force, the particle experiences a vertical force $\vec{F} = -c\hat{k}$ with $c > 0$. Let $\beta = \frac{2c\epsilon_0}{q\sigma}$. Which of the following statement(s) is(are) correct? **[JEE (Advanced) 2022]**

(A) For $\beta = \frac{1}{4}$ and $z_0 = \frac{25}{7}R$, the particle reaches the origin.

(B) For $\beta = \frac{1}{4}$ and $z_0 = \frac{3}{7}R$, the particle reaches the origin.

(C) For $\beta = \frac{1}{4}$ and $z_0 = \frac{R}{\sqrt{3}}$, the particle returns back to $z = z_0$.

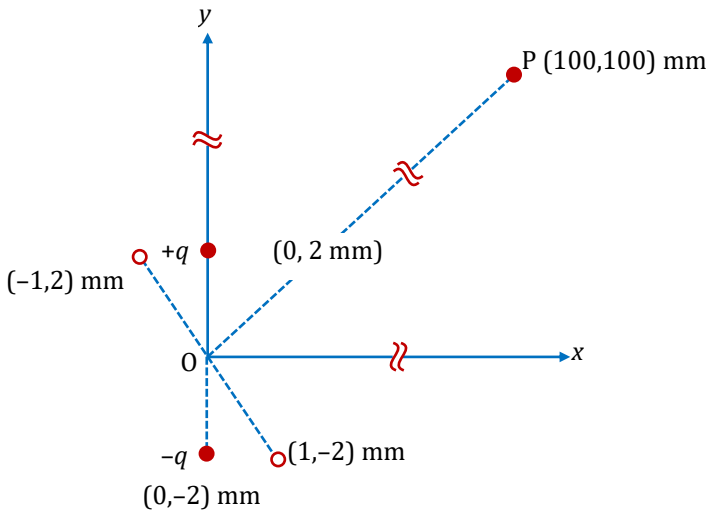
(D) For $\beta > 1$ and $z_0 > 0$, the particle always reaches the origin.

PEO075

For More Join: @IITJEE_Advanced

23. An electric dipole is formed by two charges $+q$ and $-q$ located in xy-plane at (0, 2) *mm* and (0, –2) *mm*, respectively, as shown in the figure. The electric potential at point P(100, 100) *mm* due to the dipole is V_0. The charges $+q$ and $-q$ are then moved to the points (–1, 2) *mm* and (1, –2) *mm*, respectively. What is the value of electric potential at P due to the new dipole ?

[JEE (Advanced) 2023]



(A) $V_0/4$ (B) $V_0/2$ (C) $V_o/\sqrt{2}$ (D) $3V_0/4$

PEO270

24. Two beads, each with charge q and mass m, are on a horizontal, frictionless, non-conducting, circular hoop of radius R. One of the beads is glued to the hoop at some point, while the other one performs small oscillations about its equilibrium position along the hoop. The square of the angular frequency of the small oscillations is given by [ε_0 is the permittivity of free space]

[JEE (Advanced) 2024]

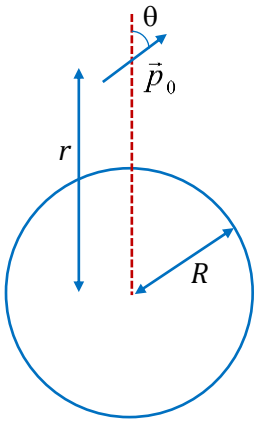
(A) $q^2/(4\pi\varepsilon_0 R^3 m)$ (B) $q^2/(32\pi\varepsilon_0 R^3 m)$ (C) $q^2/(8\pi\varepsilon_0 R^3 m)$ (D) $q^2/(16\pi\varepsilon_0 R^3 m)$

PEO309

25. A small electric dipole $\vec{p}_0$, having a moment of inertia I about its center, is kept at a distance r from the center of a spherical shell of radius R. The surface charge density σ is uniformly distributed on the spherical shell. The dipole is initially oriented at a small angle θ as shown in the figure. While staying at a distance r, the dipole is free to rotate about its center.

If released from rest, then which of the following statement(s) is(are) correct?

[JEE (Advanced) 2024]



[ε_0 is the permittivity of free space.]

(A) The dipole will undergo small oscillations at any finite value of r.

(B) The dipole will undergo small oscillations at any finite value of $r > R$.

(C) The dipole will undergo small oscillations with an angular frequency of $\sqrt{\frac{2\sigma p_0}{\epsilon_0 I}}$ at $r = 2R$.

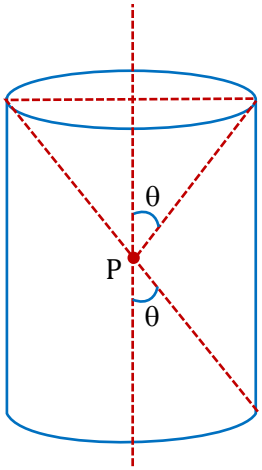
(D) The dipole will undergo small oscillations with an angular frequency of $\sqrt{\frac{\sigma p_0}{100\,\epsilon_0 I}}$ at $r = 10R$.

PEO310

For More Join: @IITJEE_Advanced

26. A charge is kept at the central point P of a cylindrical region. The two edges subtend a half-angle θ at P, as shown in the figure. When $\theta = 30°$, then the electric flux through the curved surface of the cylinder is Φ. If $\theta = 60°$, then the electric flux through the curved surface becomes $\Phi/\sqrt{n}$, where the value of n is_____.

[JEE (Advanced) 2024]

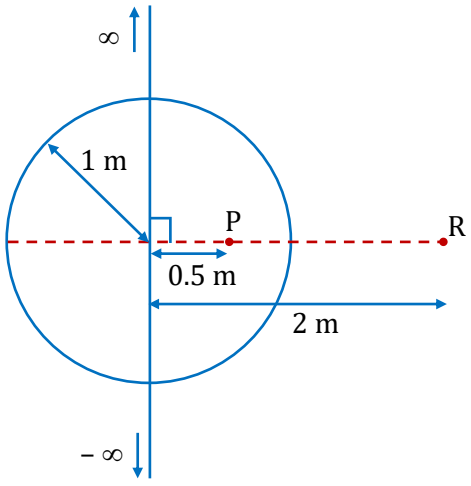


PEO311

27. An infinitely long thin wire, having a uniform charge density per unit length of 5 nC/m, is passing through a spherical shell of radius 1 m, as shown in the figure. A 10 nC charge is distributed uniformly over the spherical shell. If the configuration of the charges remains static, the magnitude of the potential difference between points P and R, in Volt, is _____.

[Given: In SI units $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$, $ln\ 2 = 0.7$. Ignore the area pierced by the wire.]

[JEE (Advanced) 2024]



PEO312

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	C	C	D	A	C	B	B	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	B	D	B	D	B	A	C	D	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	C	A	C	B	A	B	B	C	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	B	D	A	C	D	A	D	C	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	B	B	A	A	B	B	B	D	D	C
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	A	C	B	C	B	A	D	C	A	C
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	B	A	A	C	B	B	D	B	C	D
Que.	71	72	73	74	75	76	77	78	79	80
Ans.	D	B	B	B	C	B	A	B	A	C
Que.	81	82	83	84	85	86	87	88	89	90
Ans.	A	A	B	C	C	A	C	A	A	D
Que.	91	92	93	94	95	96	97	98	99	100
Ans.	D	C	A	C	A	D	B	D	C	A
Que.	101	102	103	104	105	106				
Ans.	D	A	A	B	C	A				

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EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ABC	ABC	BC	AB	AD	ABC	BC	BD	B	A
Que.	11	12	13	14	15	16				
Ans.	C	B	D	A	A	(A-Q,R), (B-Q,T), (C-S,T), (D-P)				

EXERCISE (S)

1.	625	**2.**	$\frac{9F}{4}$	**3.**	3	**4.**	9
5.	3	**6.**	8	**7.**	400	**8.**	3
9.	2	**10.**	1	**11.**	2	**12.**	12
13.	4	**14.**	6	**15.**	3	**16.**	576
17.	1	**18.**	8	**19.**	3	**20.**	2
21.	6	**22.**	2	**23.**	20	**24.**	2
25.	2	**26.**	2	**27.**	1	**28.**	2
29.	6	**30.**	4				

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	1	4	1	4	2	4	1	–48.00	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	3	300	8	2	1	4	27	4	3

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	6	C	BD	AB	AB	2 [1.99, 2.01]	B	A	ABD
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	AD	BC	3.14	6.40	2	BC	1.73	3	ABC	3
Que.	21	22	23	24	25	26	27			
Ans.	B	ACD	B	B	BD	3	171			

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